

# CAUSAL INFERENCE TOOL KITS (WITH OBSERVATIONAL DATA OR QUASI-EXPERIMENTAL DATA)

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# AGENDA

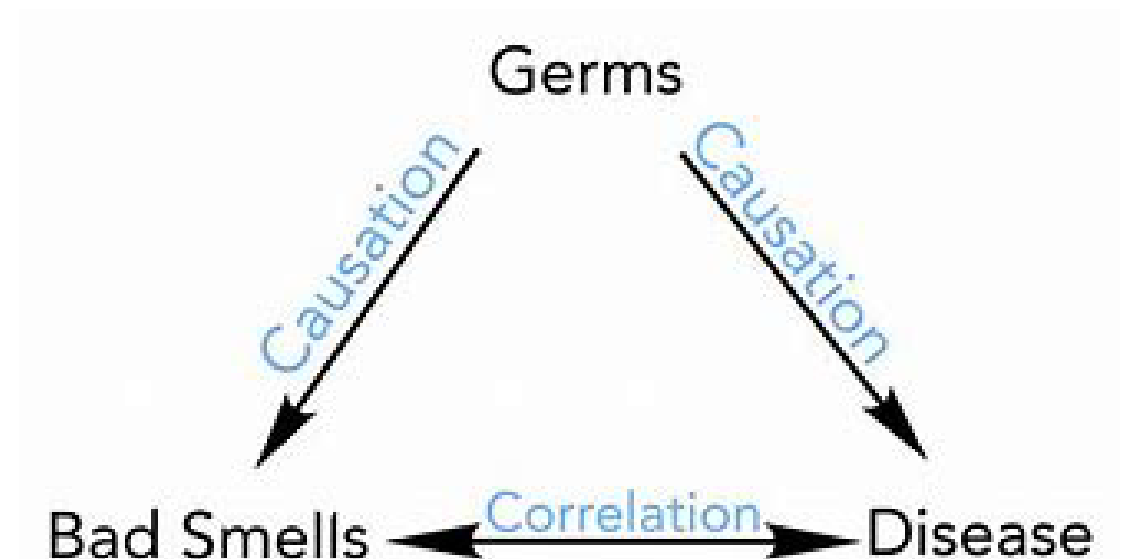
## Causal Inference

- Learning Paradigms
  - Prediction vs. Causal Inference
  - Generative AI
  - Reinforcement Learning
- Causal Inference: Two Traditions
- Typical Problems in Causal Inference with Observational Data
- Two Approaches in Econometrics
- Data Types: Cross-Section, Time Series, Panel

## Causal Inference Tool Kits

- Matching
- Propensity score matching
- Instrumental Variables
- Difference in Differences (DID)
- Regression Discontinuity
- Double/Debiased Machine Learning
- Heterogeneous Treatment Effect

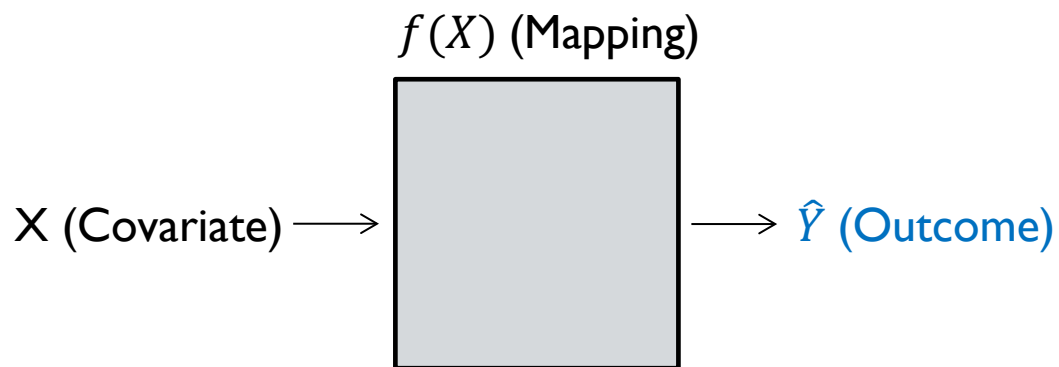
## Causal Demand Bench



# Prediction vs. Causal Inference

## Prediction: Traditional ML (Supervised Learning)

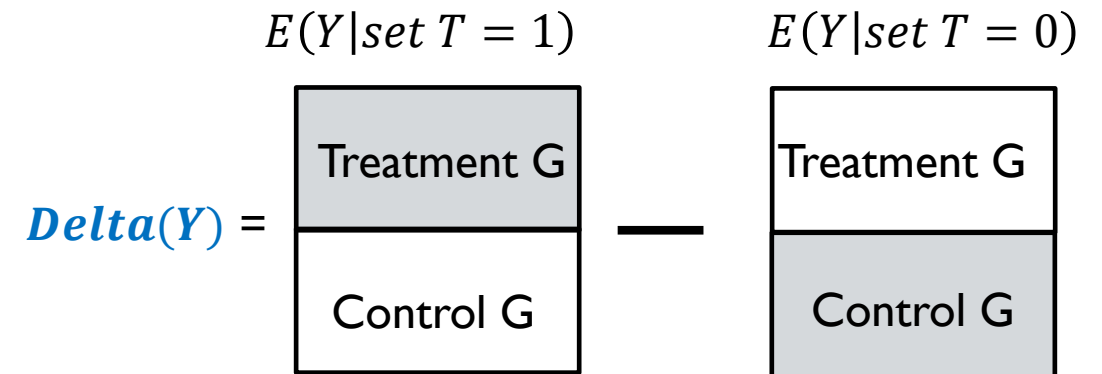
- Predict Y after **observing  $X=x$**
- **Flexible functional mapping** from X to Y
  - $E(Y|observe\ X) = f(X)$
- **Black box**
- **Correlational**



How many users will use ChatGPT in Aug 2026?

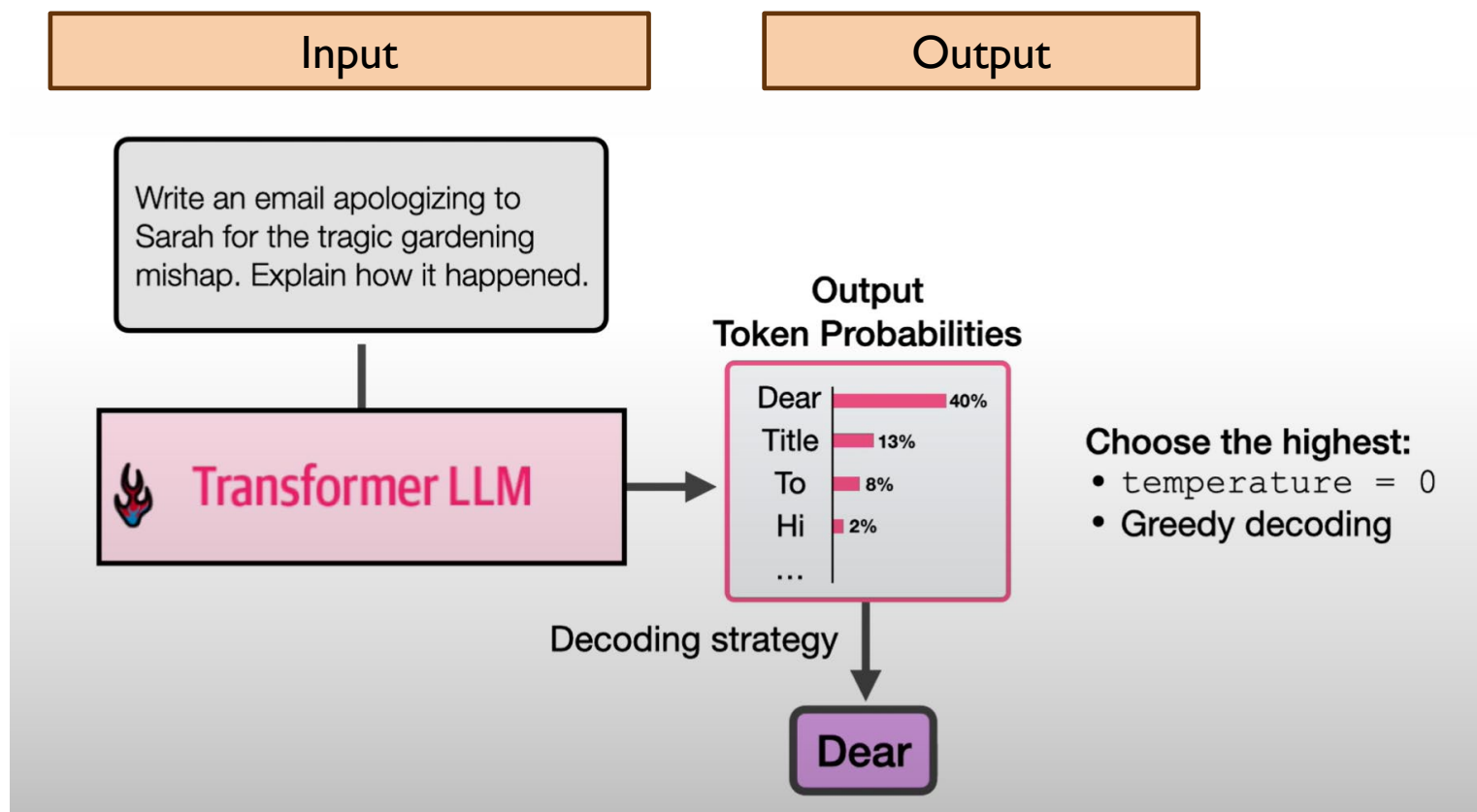
## Causal Inference: Causal ML

- Predict effect on Y after **setting  $X=x$**
- **Counterfactuals** based on potential outcomes
  - $\Delta$  (Effect)  
 $= E(Y|set\ T = 1) - E(Y|set\ T = 0)$
- **Interpretable**
- **Causal** **Missing Data: Potential Outcome**



How many more users (i.e., incremental users) are using ChatGPT because of new Cowork feature in Aug 2026?

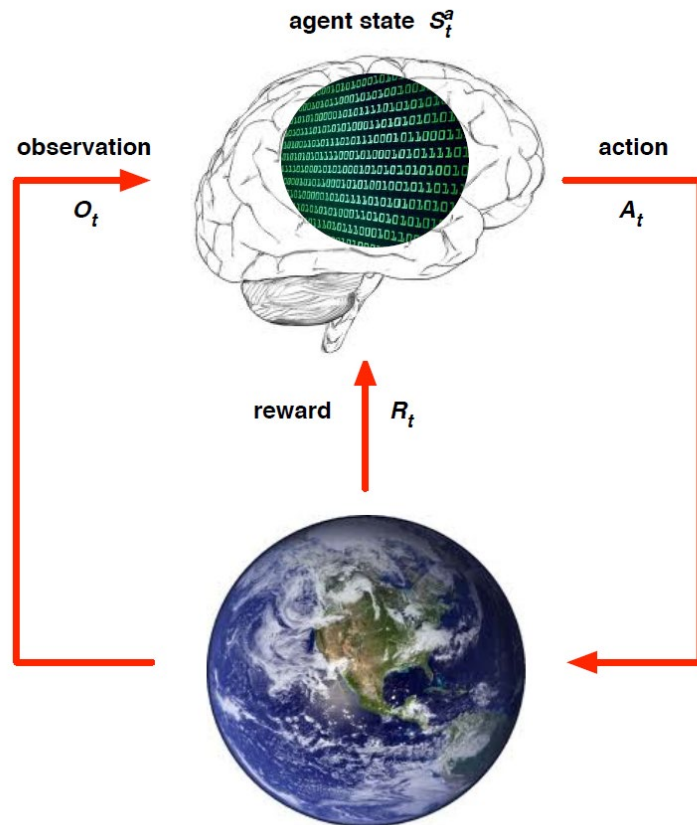
# Generative AI – Autoregressive Transformer LLM



A language model is a **probability kernel  $\mu$**  given a prefix of words:  **$\mu: X \Rightarrow Pr(Y)$**

Traditional ML: **deterministic output (not distribution)** – classification or regression

# Reinforcement Learning: Agent and Environment



- At each step  $t$ , **the agent**:
  - Execute action  $A(t)$
  - Receives observation  $O(t)$
  - Receives scalar reward  $R(t)$
- The **environment**:
  - Receives action  $A(t)$
  - Emits observation  $O(t+1)$
  - Emits scalar reward  $R(t+1)$
- **$t$  increment** at environment step

- **Sequential Decision Making, (Non-Parametric) Stochastic Optimization**
- Reward Hypothesis: All goals can be described by the maximization of **expected cumulative reward (scalar)**

# Causal Inference Frameworks

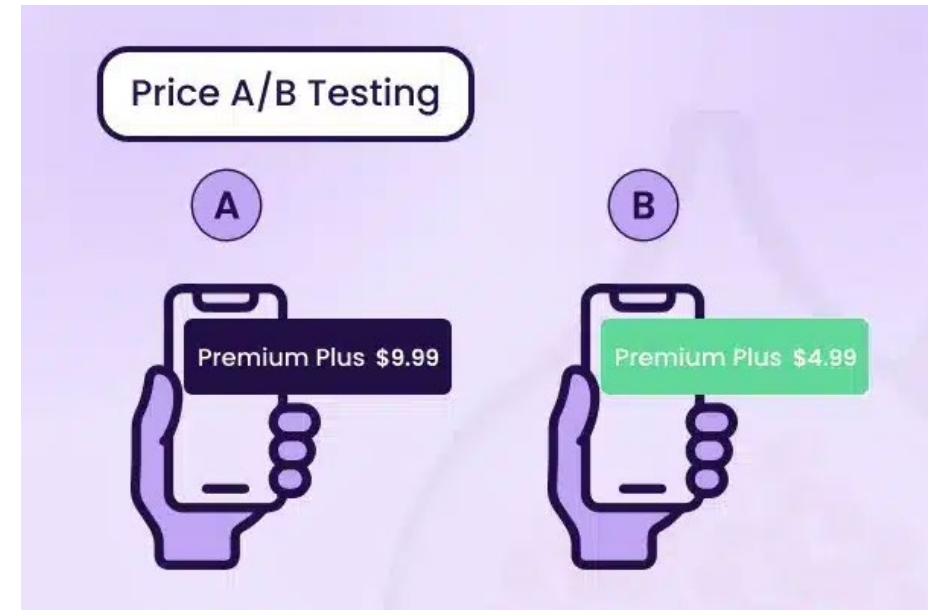
Focus Today

|                  | Causal Discovery (CD)                                                          | Causal Representation Learning (CRL)                           | Causal Effect Estimation (CEE)                                              |
|------------------|--------------------------------------------------------------------------------|----------------------------------------------------------------|-----------------------------------------------------------------------------|
| Question?        | What cause what?                                                               | What are the hidden causes behind raw data?                    | How much does change in X cause change in Y?                                |
| Primary Goal     | Map the causal relationships between observed variables                        | Extract latent variables and their graph from complex raw data | Measurement and Control, Quantify the effect size from certain intervention |
| Input Data Type  | Structured, low-dimensional tabular data                                       | Unstructured, high-dimensional raw data (pixels, text, audio)  | Structured tabular data                                                     |
| Variable State   | Fully observed (predefined by humans)                                          | Hidden/Latent (must be learned by AI)                          | Fully observed (causal graph must be known beforehand)                      |
| Typical Problems | Unobserved confounders are not handled, Disentangle causality from correlation | Non-identifiability                                            | Adjusting for confounding (e.g., endogeneity) and selection bias            |

# Causal Effect Estimation: (I) Randomized Experiment

## Pricing Experiments: A/B tests

- High **opportunity cost: lost profits**
- Extremely **large action space** with many products
- **Coupled rewards**: cross price elasticities
- **Regulatory constraint**: no price discrimination across customers for the same product and same quantity

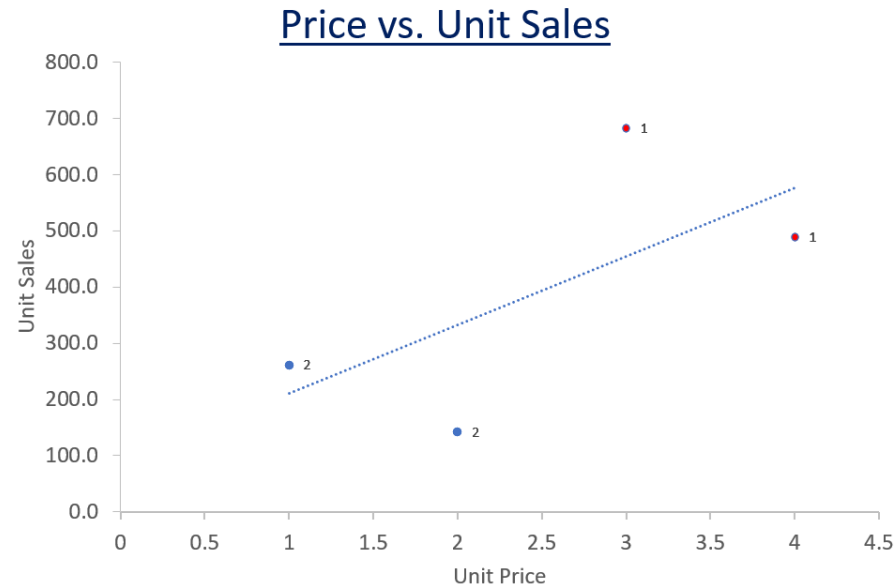


Practically, only feasible for **single product or a small subset of products**: e.g., multi-armed bandit, contextual bandit

# Causal Effect Estimation: (2) Observational Causal Inference

Exploring historical offline data: Causal demand estimation

- **Endogeneity**: Lack of exogenous variation
- **“Greedy method”** – no exploration (on-policy)



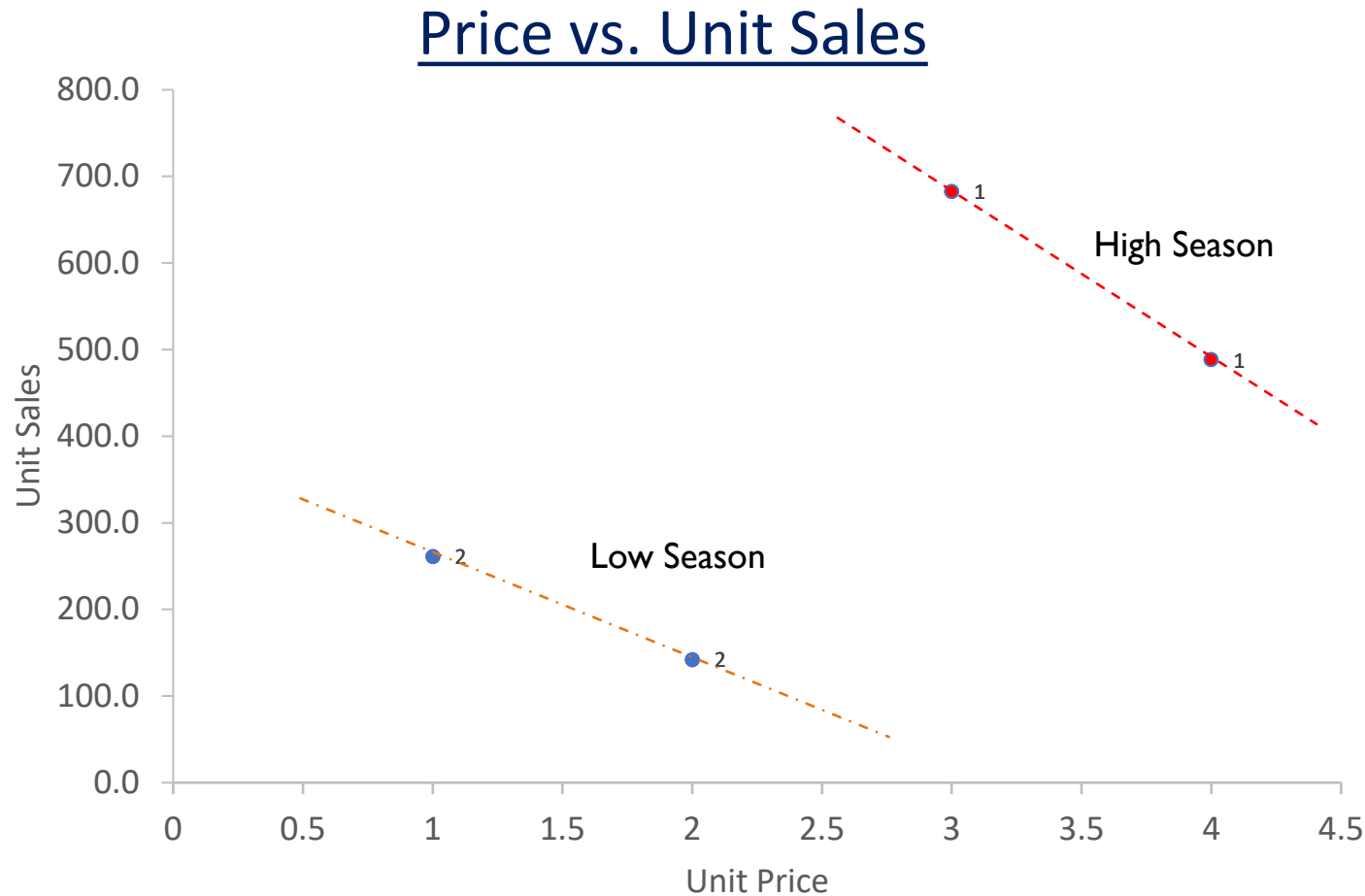
- **Positive Price Elasticity?**
- **Does sales go up with increasing price?**

EXISTING APPROACHES SUFFER FROM A LACK OF STRATEGIC EXPLORATION



## Causal Effect Estimation: (2) Observational Causal Inference

### - Challenge: Problem with Unobserved Seasonal Demand



- True: Negative Price Elasticity
- Confounding from high/low season: High demand season has higher prices and sales

# Two Traditions in Causal Inference

## Potential Outcomes (Neyman-Rubin)

- Computational Social Science: Econometrics, Marketing Science, Operation Research, Finance, Political Science, Clinical Studies
  - [Prof. Guido Imbens \(Stanford\)\\*](#): Econometric Method
  - [Prof. John Angrist \(MIT\)\\*](#): Labor Economics
  - [Prof. David Card \(UC Berkeley\)\\*](#): Labor Economics (e.g., Minimum wage tax)
  - [Prof. Susan Athey \(Stanford\)](#): Econometric Method
- Books and Blogs
  - [Statistics and Causal Inference \(harvard.edu\)](#)
  - [Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction: Imbens, Guido W., Rubin, Donald B.](#)
  - [Mostly Harmless Econometrics](#)
  - [Causal Inference for The Brave and True — Causal Inference for the Brave and True \(matheusfacure.github.io\)](#)
  - [Causal Machine Learning Archives - Microsoft Research \(Cambridge\)](#)
- Open-Source Package
  - [ECON ML](#)
  - [Causal ML](#)

## Graph (Structural Equation Model)

- Computer Science (Causal ML, Bayesian Network), Marketing Science (Structural Equation Model, Bayesian Network)
  - [Prof. Judea Pearl \(UCLA\)\\*\\*](#)
  - [Silvia Chiappa \(DeepMind\)](#)
- Books and Blogs
  - [Causal Inference in Statistics: A Primer](#)
  - [Causality: Models, Reasoning, and Inference](#)
  - [The Book of Why: The New Science of Cause and Effect](#)
  - [Project Causica: Decision Optimization with Causal ML - Microsoft Research \(London\)](#)
- Open-Source Package
  - [Do Why](#)
  - [Show Why](#)
  - [Causica](#)

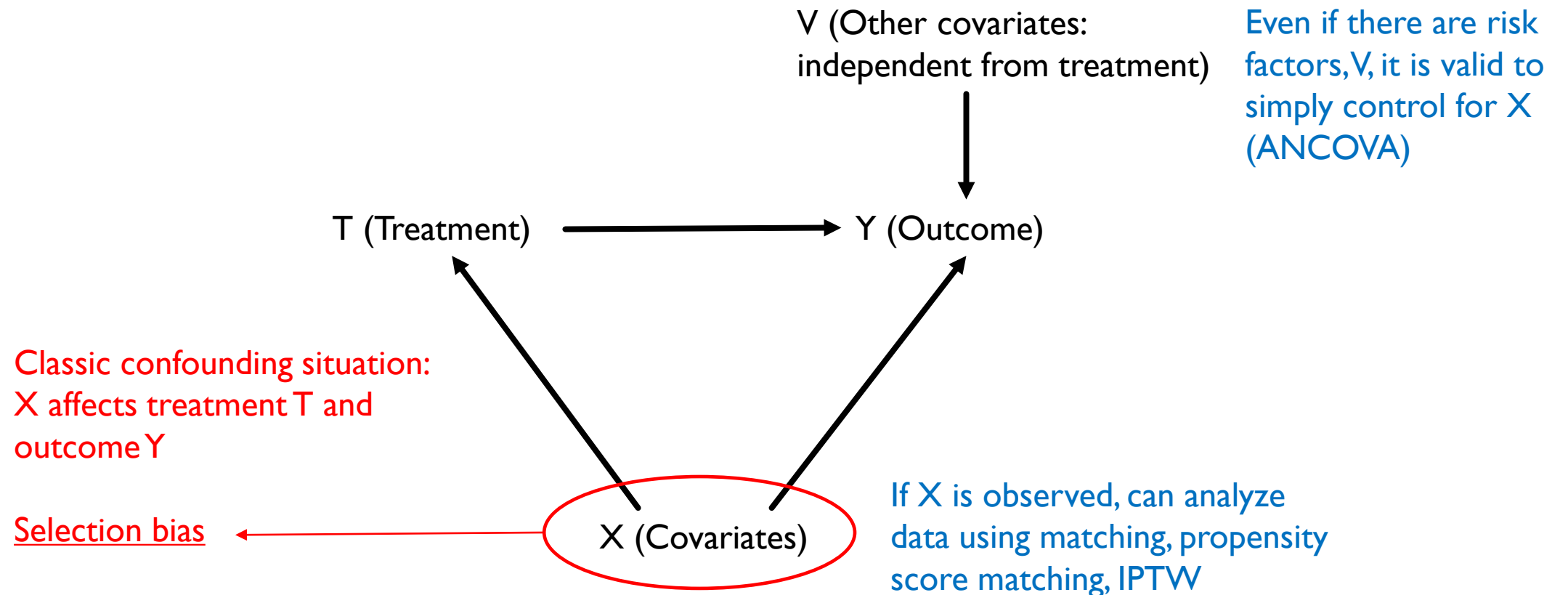
[Causal Data Science Meeting 2026](#)

\* 2021 Economics Nobel Prize Winner

\*\* 2011 ACM Turing Award Winner

# Typical Problems Faced with Causal Inference with Observational Data (1/2)

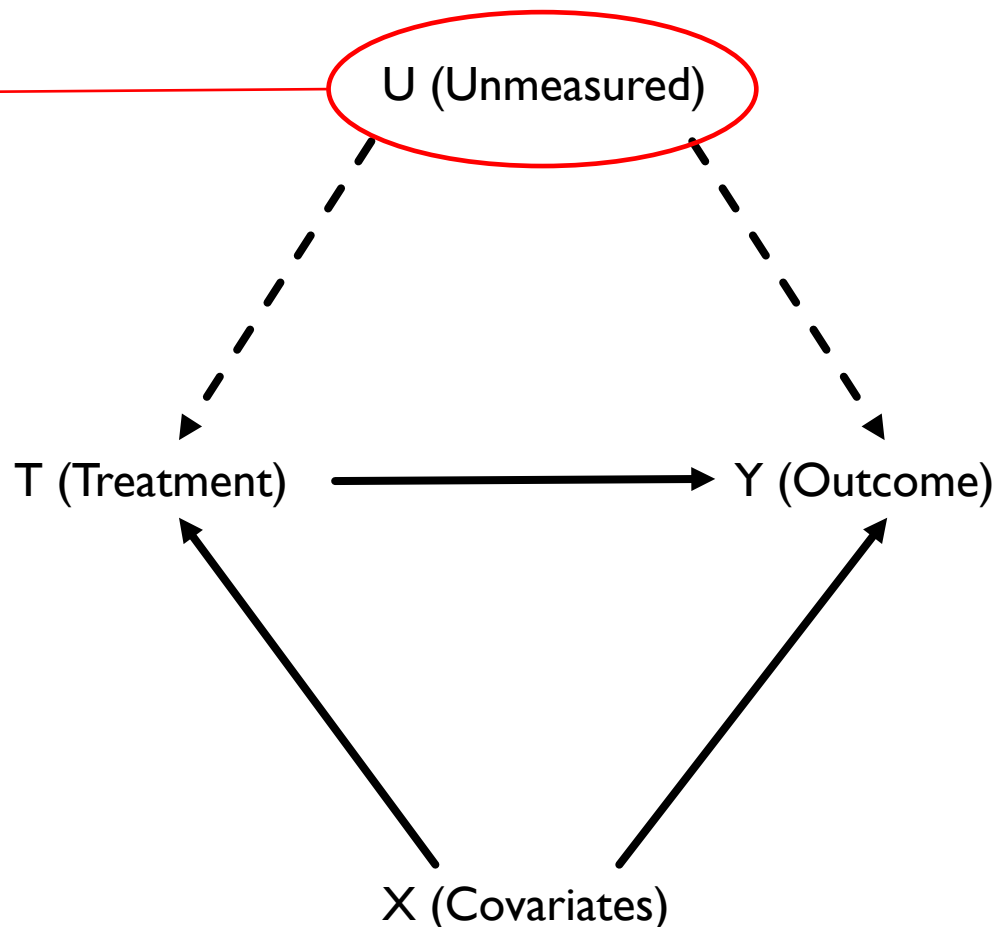
Confounding from observed covariates: **matching, propensity score matching, IPTW**



## Typical Problems Faced with Causal Inference with Observational Data (2/2)

Unmeasured confounding: **instrumental variable** to the rescue

Omitted variable bias  
(endogeneity)

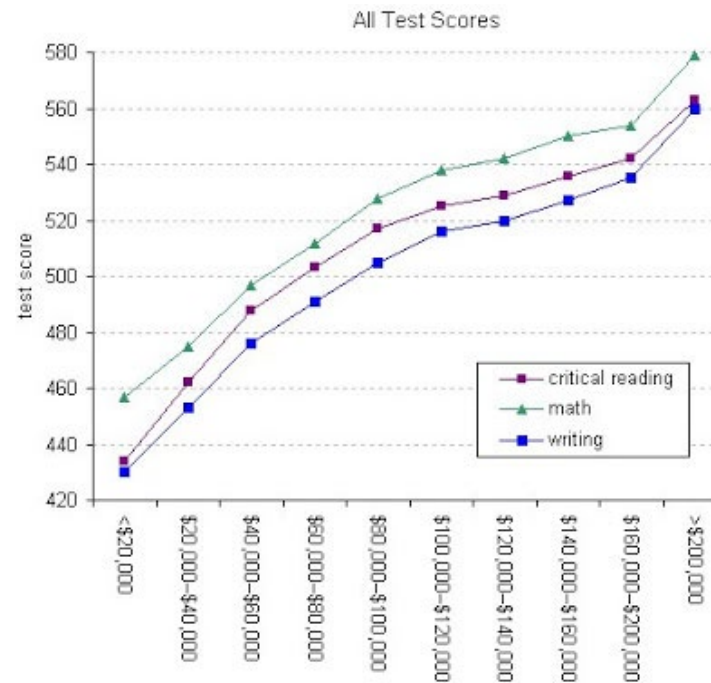


- Ignorability assumption\* violated
- Biased estimates of causal effects
- We **cannot control for confounders** and average over the distribution if we do not observe them all

\* Condition on X, confounding can be ignored

# Confounders can be dangerous

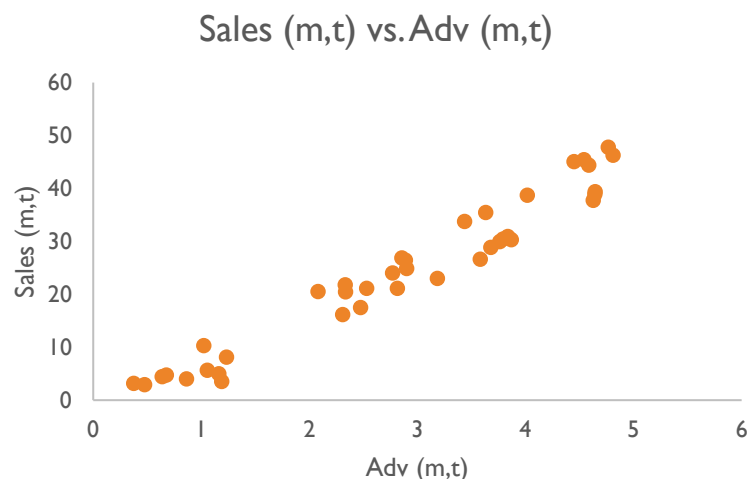
- Oftentimes, we don't have data on some important variables, which can impact sales, revenue, or profits.
  - Doing analytics ignoring these “omitted variables” can lead to “biased” estimates of promotion effects and marketing mix effects.
- Think about the graph below (from NYT). Is the family income really responsible for better academic achievement? What would be potential omitted variable bias here?



**Failing to control confounders can lead to false conclusions**

# Simultaneity Bias: Careful with Reverse Causality

- Imagine the following data generating process.
  - $Sales_{m,t} = Intercept_m + 0 \cdot Adv_{m,t} + \varepsilon_{m,t}$  (True advertising effect: 0)
  - $Adv_{m,t} = 0.1 \cdot Sales_{m,t-1} + \omega_{m,t}$
- If you do analysis ignoring the reverse causality, you may conclude the following.



|                 | Coefficients | Standard Error | P-value |
|-----------------|--------------|----------------|---------|
| Intercept       | -3.36        | 1.12           | 0.004   |
| Advertising (t) | 9.72         | 0.35           | 0.000   |

Significant impact of advertising?

R square: 0.95 (Good fit?)

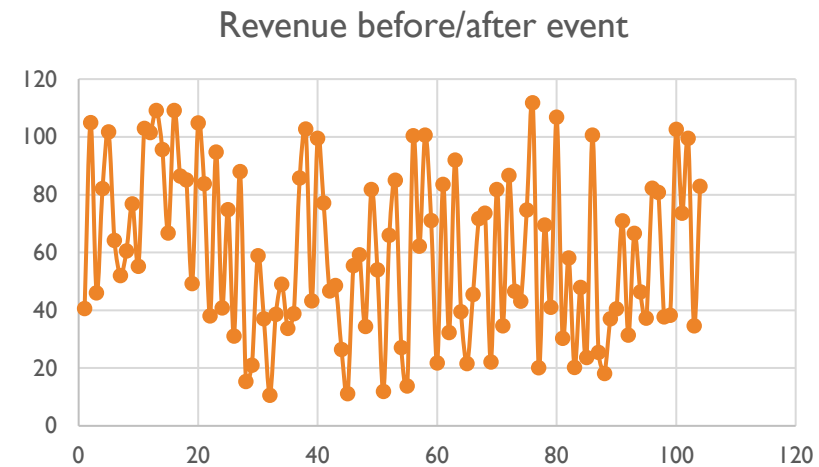
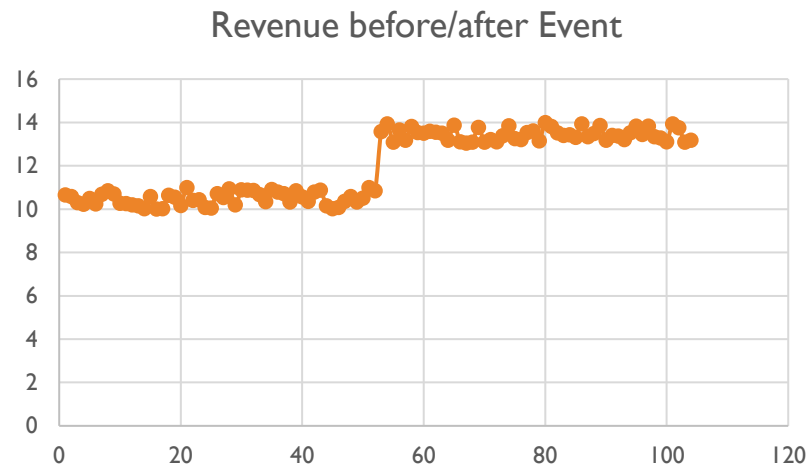
**Need to understand advertising assignment rules. (Correlation != Causality)**

# Pay attention to signal-to-noise ratio!

- Consider the following measurement. Is there significant impact from marketing event?

| Revenue | Before Event | After Event | % Change |
|---------|--------------|-------------|----------|
| Average | 10           | 13          | 30%      |

- Well, it depends on signal-to-noise ratio!



**Measuring means is not enough. Pay attention to variance, t-statistics, and confidence intervals**

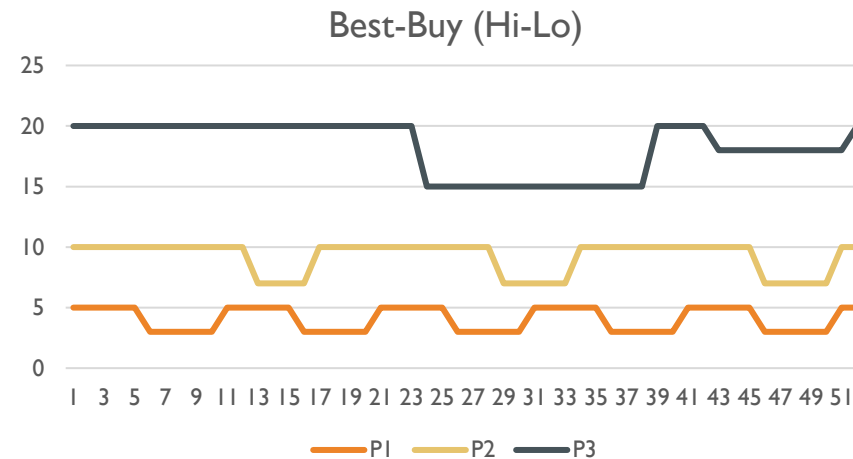
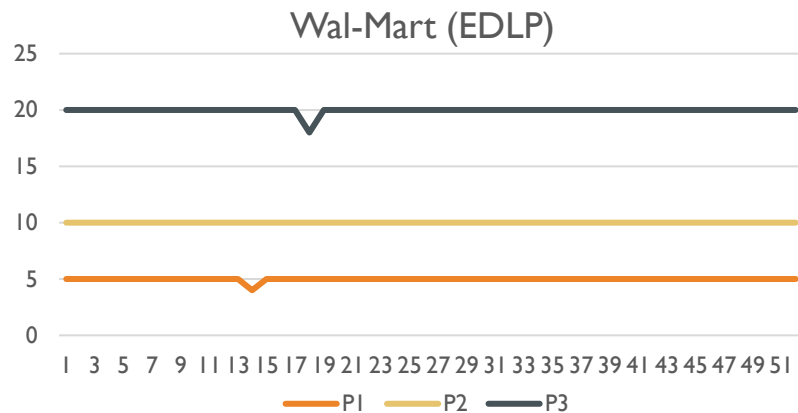
# Size of Data $\neq$ Information in Data

- Consider the following two options:

(1) Wal-Mart with 4,000 stores, 52 weeks of data, 500 SKUs (104 million observations!)

(2) Best-buy with 1,500 stores, 52 weeks of data, 500 SKUs (39 million observations)

- Which dataset would be more useful to measure price responses?



**Check the variation of prices or promotion tactics (e.g. coefficient of variation)**



# Two Approaches in Econometrics

## Reduced-Form Causal

- **Data-Centric**
  - Do not make strong assumption for generative model (i.e., functional form)
  - Utilizes natural experiment in the data (i.e., quasi-experiment)
  - Requires deep understanding of data generating process, business or policy context: e.g., how advertising budget is set, how personalized discount amount is set, how tax credit is set
- **Some assumptions (weaker, partly testable) + Focuses of causal effect measurement** (can't extrapolate data range)
- **Typical approaches:**
  - Matching
  - Propensity score matching (and IPTW)
  - Instrumental variables
  - Difference in differences
  - Regression discontinuity
  - Synthetic control
  - Double ML: Meta Learner, Uplift Modeling

## Structural Econometrics

- **Data + Theoretical Structure:** Complement data with theory-based structure
- Make **strong assumption; Extrapolate beyond data range based on economic or psychological theories (i.e., counterfactual simulation)**
- **Typical approaches:**
  - Utility maximization: Discrete choice model of consumer behavior (Daniel McFadden, Nobel Prize 2000)
  - Markov decision process: Dynamic discrete choice, Value function, policy function
  - Nash equilibrium: Static and Dynamic game models (Static Entry Game, Dynamic Entry Game, Innovator's Dilemma)
  - Aggregation of simulated individuals: BLP demand model
  - Weizmann Sequential Search Model: e.g., [searchChoiceProbit.pdf](#)
  - Matching Game
  - Auction
  - Social interaction
  - Heuristics and Biases (Behavioral decision theory): e.g., Prospect Theory

# Data types and source of variation

## Cross-section

- Multiple (cross-sectional) units:  $N$
- Single point in time:  $T = 1$
- Large  $N$ , 1-dimensional  $T$  problem
- Source of variation: across units (i.e., **between** variation)

## Time Series

- Single unit:  $N = 1$
- Multiple points in time:  $T$
- 1-dimensional  $N$ , large  $T$  problem
- Source of variation: within a unit, across time (i.e., **within** variation)

## Panel (Longitudinal)

- Multiple units:  $N$
- Multiple points in time:  $T$
- Most likely: large  $N$ , small  $T$  problem
- Source of variation: within a unit, across time (i.e., **within** variation) + across units (i.e., **between** variation)

# Matching

- Solving confounding from observed variables (i.e., selection bias)
- With confounder X, treated and untreated are initially not comparable: covariate imbalance
- Match each treated unit with a similar untreated unit
  - Potential curse of dimensionality: sometimes it is difficult to find great matches
  - Willing to accept some non-ideal matches if treated and control groups have same distribution of covariates

Example with 3 covariates: Age, Chronic lung disease (1=Yes, 0=No), Female (1=Yes, 0=No)

## Treated

| Age   | Chronic lung disease | Female |
|-------|----------------------|--------|
| 78.17 | 0                    | 1      |

## Control

| Age   | Chronic lung disease | Female | Distance |
|-------|----------------------|--------|----------|
| 75.33 | 0                    | 1      | 0.17     |
| 54.97 | 0                    | 0      | 2.45     |
| 18.04 | 0                    | 1      | 3.60     |
| 70.25 | 1                    | 0      | 4.23     |

# Number of matches

- One to one (pair matching)
  - Match exactly one control to every treated unit
- Many to one
  - Match some fixed number  $K$  controls to every treated subjects (e.g., 5 to 1 matching)
- Variable
  - Sometimes match 1, sometimes more than 1 control to treated units
  - If multiple good matches available, use them. If not, do not.

# Matching: distance metric

Because we typically cannot match exactly, we need to choose some metric of closeness

- Two options
  - Mahalanobis distance (between covariates of unit  $i$  and unit  $j$ ): square root of the sum of squared distances between each covariate by the covariance matrix

$$D(X_i, X_j) = \sqrt{(X_i - X_j)^T S^{-1} (X_i - X_j)}$$

$X_i$ : a vector of covariates for unit  $i$

$S^{-1}$ : covariance for scaling (for differences in covariate measurement unit)

- Robust Mahalanobis distance: Outlier-robust
  - Rosenbaum, P. R., “Design of Observational Studies”, Springer Science & Business Media (2009)
  - (1) Replace each covariate value with its rank
  - (2) Constant diagonal on covariance matrix (Ranks should be on the same scale)
  - (3) Calculate the usual Mahalanobis distance on the ranks

# How to match? (1/3)

Once you have distance scores, how to select matches?

- Usually, you have **more controls** than treated units
- Focus on pair (one-to-one) matching (for now)
- 2 options: **Greedy** vs. **Optimal**

## How to match? (2/3): Greedy matching

- **Greedy** (nearest neighbor) matching: One at a time
  - **Not as good**, but **computationally fast** (for large dataset)
  - Steps)
    1. Randomly order list of treated units and control units
    2. Start with the first treated unit. Match to the control with the smallest distance.
    3. Remove the matched control from the list of available matches
    4. Move on to the next treated unit. Match to the control with the smallest distance
    5. Repeat steps 3 and 4 until you have matched all treated units.
  - **R packages; MatchIt** (No good Python packages yet: You can use ryp2 package to run R in Python)

(For K:1 matching – many-to-one, after everyone has 1 match, go through the list again and find 2nd matches.  
Repeat until k matches

- **Caliper**: We might prefer to exclude treated units for whom there does not exist a good match
  - A bad match can be defined using a **caliper** – maximum acceptable distance.
  - Only match a treat unit if the best match has distance less than the caliper.

- **Sample code**

```
#do greedy matching on Mahalanobis distance
greedymatch<-Match(Tr=treatment,M=1,X=mydata[xvars])
matched<-mydata[unlist(greedymatch[c("index.treated","index.control")])), ]
```

## How to match? (3/3): Optimal matching

- Optimal matching: in terms of global distance
  - Better, but computationally demanding: 100 treated units and 1000 controls results in 100,000 possible pairings (1 million treatment-control pairing is feasible, 1 billion pairings is likely not feasible.)
  - Minimizes global distance measure (e.g., total distance)
  - Consider all possible matches between treatment and control
  - Can be solved with network flow optimization problem
  - R packages: optmatch, rcbalance



# Checking Covariate Balance: TableOne

- **TableOne** analysis is useful to check whether confounding (from observed covariate imbalance is potentially a problem
  - Often, **standardized differences** are displayed in table, both before and after matching

$$smd = \frac{\bar{X}_{treatment} - \bar{X}_{control}}{\sqrt{\frac{s_{treatment}^2 + s_{control}^2}{2}}}$$

- Rules of thumb:
  - Values < 0.1: indicate adequate balance
  - Values 0.1 – 0.2: are not too alarming
  - Values > 0.2: indicate serious imbalance
- Code sample

```
#look at a table 1
table1<- CreateTableOne(vars=xvars,strata="treatment", data=mydata, test=FALSE)
## include standardized mean difference (SMD)
print(table1,smd=TRUE)
```

## Checking Covariate Balance: TableOne - Example

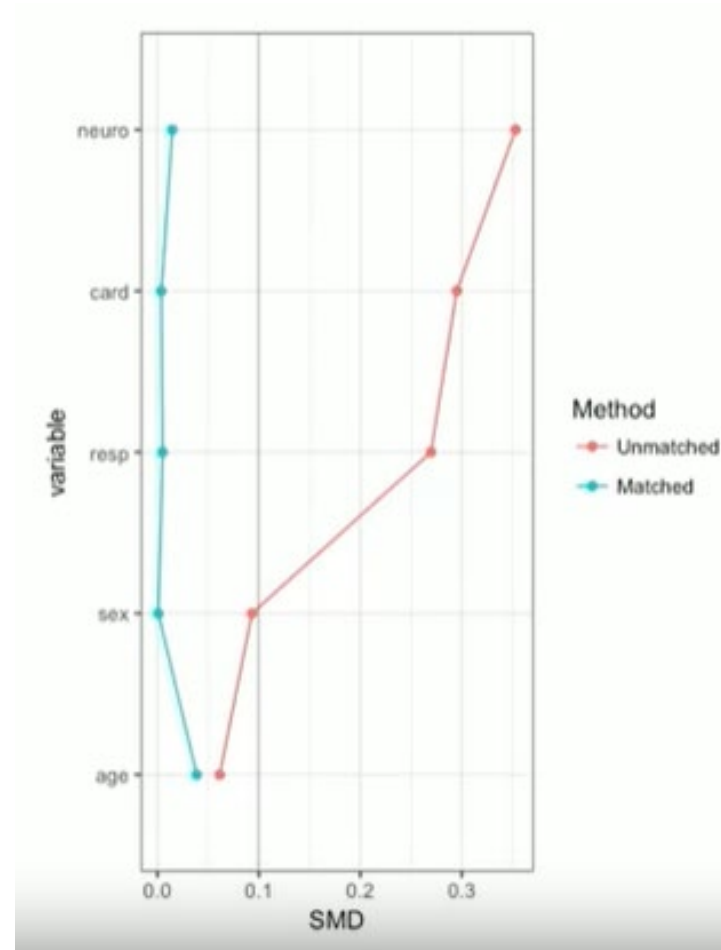
- Right heart catheterization (RHC) data

 Covariate imbalance

|                 | Unmatched   |             |      | Matched     |             |       |
|-----------------|-------------|-------------|------|-------------|-------------|-------|
|                 | No RHC      | RHC         | SMD  | No RHC      | RHC         | SMD   |
| n               | 3551        | 2184        |      | 2082        | 2082        |       |
| age (mean (sd)) | 61.8 (17.3) | 60.8 (15.6) | 0.06 | 61.6 (16.7) | 61.0 (15.8) | 0.039 |
| sex = Male (%)  | 53.9        | 58.5        | 0.09 | 56.9        | 56.9        | 0.001 |
| resp = Yes (%)  | 41.7        | 28.9        | 0.27 | 30.6        | 30.4        | 0.005 |
| card = Yes (%)  | 28.4        | 42.3        | 0.30 | 39.3        | 39.5        | 0.004 |
| neuro = Yes (%) | 16.2        | 5.4         | 0.35 | 5.3         | 5.7         | 0.015 |

# Checking Covariate Balance: SMD Plot - Example

- Right heart catheterization (RHC) data



# Analysis after matching (1/2): Permutation test (randomization test)

- Permutation tests
  - Compute **test statistic** from observed data (after matching)
  - Assume null hypothesis of **no treatment effect** is true
  - Randomly **permute treatment assignment within pairs** and re-compute test statistic
  - Repeat many times and see **how unusual** observed statistic is

## Analysis after matching (2/2): Paired Test

- Binomial data (0/1)
  - Permutation test is equivalent to the [McNemar test](#) for paired binomial data

```
> ex<-matrix(c(1,5,2,5),2,2)
> mcnemar.test(ex)

McNemar's Chi-squared test with continuity correction

data:  ex
McNemar's chi-squared = 0.57143, df = 1, p-value = 0.4497
```

|           |         | Control |   |
|-----------|---------|---------|---|
|           |         | Outcome |   |
|           |         | 1       | 0 |
| Treatment | Outcome |         |   |
|           | 1       | 1       | 5 |
|           | 0       | 2       | 5 |

- Continuous data
  - Could also use a [paired t-test](#)

```
> t.test(txsbp, consbp, paired=TRUE)

Paired t-test

data:  txsbp and consbp
t = -6.7416, df = 19, p-value = 1.928e-06
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -10.749497 -5.656137
sample estimates:
mean of the differences
      -8.202817
```

# Propensity score matching avoids curse of dimensionality

- The propensity score is the **probability of receiving treatment**, rather than control, given covariate  $X$
- Define  $T=1$  for treatment and  $T=0$  for control
- The **propensity score for unit  $i$** :

$$\pi_i = P(T = 1|X_i)$$

- The propensity score is **balancing score**
  - Suppose that 2 units have the same value of the propensity score, but they possibly have different covariate values  $X$
  - Despite the different covariate values, they were both equally likely to have been treated
  - This means that both units'  $X$  is just as likely to be found in the treatment group
  - Propensity score matching will ensure that **balances are achieved in distribution of covariates**

$$P(X = x|\pi(X) = p, T = 1) = P(X = x|\pi(X) = p, T = 0)$$

# Estimating propensity score

- We need to estimate  $P(T = 1|X)$
- The ‘outcome’ here is T (assignment into treatment), which is binary
- We can use binary logistic regression\*
  - Fit a logistic regression model: outcome T, covariate X
  - From that model, get the predicted probability (fitted value) for each unit. That is the estimated propensity score
  - If assignment was done by “unobserved” covariate(s), we can’t properly estimate this.

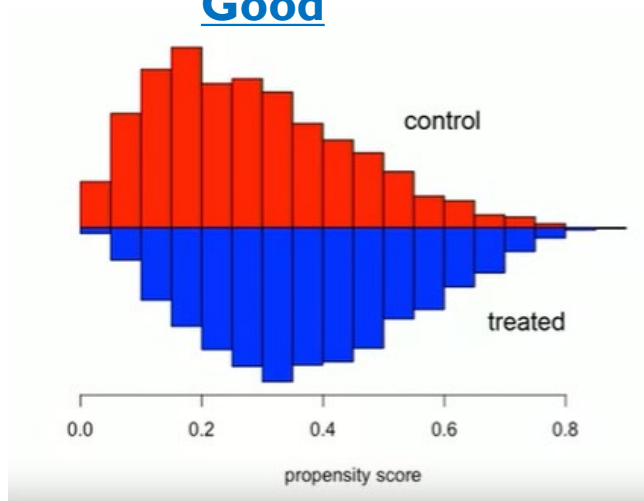
\* Other methods for estimating a propensity score could be used, including machine learning methods

Source: [A Crash Course in Causality: Inferring Causal Effects from Observational Data - Home | Coursera](#)

# Overlap: Useful to look before propensity score matching

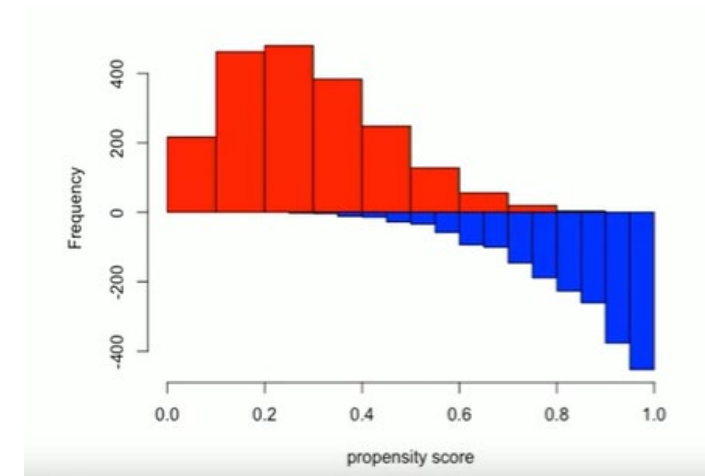
- Compare the distribution of the propensity score for treated and control units
  - Often done with a plot
- Good vs. bad **overlap** examples

**Good**



- **Good overlap everywhere**
- **Positivity assumption** seems **reasonable**: all units have a chance to receive a treatment

**Bad**



- **Poor overlap**
- **Positivity assumption** likely **violated**
- Trimming the tail is an option



## Example in R: Fit propensity score model\* (1/2)

```
#fit a propensity score model. logistic regression

psmodel<-glm(treatment~ARF+CHF+Cirr+colcan+Coma+lungcan+MOSF+
             sepsis+age+female+meanbp1+aps,
             family=binomial(),data=mydata)

#show coefficients etc
summary(psmodel)
#create propensity score
pscore<-psmodel$fitted.values
```

\* Other methods for estimating a propensity score could be used, including machine learning methods

Source: [A Crash Course in Causality: Inferring Causal Effects from Observational Data - Home | Coursera](#)

## Example in R: Fit propensity score model\* (2/2)

Coefficients:

|             | Estimate   | Std. Error | z value | Pr(> z ) |     |
|-------------|------------|------------|---------|----------|-----|
| (Intercept) | -1.9460154 | 0.2321291  | -8.383  | < 2e-16  | *** |
| ARF         | 1.2252930  | 0.1495511  | 8.193   | 2.54e-16 | *** |
| CHF         | 1.8905642  | 0.1735687  | 10.892  | < 2e-16  | *** |
| Cirr        | 0.4334062  | 0.2203366  | 1.967   | 0.04918  | *   |
| colcan      | 0.0481566  | 1.1242894  | 0.043   | 0.96583  |     |
| Coma        | 0.6842545  | 0.1878333  | 3.643   | 0.00027  | *** |
| lungcan     | 0.1984600  | 0.5055005  | 0.393   | 0.69461  |     |
| MOSF        | 1.0177797  | 0.1807159  | 5.632   | 1.78e-08 | *** |
| sepsis      | 1.8402456  | 0.1561589  | 11.784  | < 2e-16  | *** |
| age         | -0.0030469 | 0.0017462  | -1.745  | 0.08101  | .   |
| female      | -0.1390768 | 0.0590139  | -2.357  | 0.01844  | *   |
| meanbp1     | -0.0075166 | 0.0008707  | -8.633  | < 2e-16  | *** |
| aps         | 0.0182356  | 0.0017286  | 10.549  | < 2e-16  | *** |

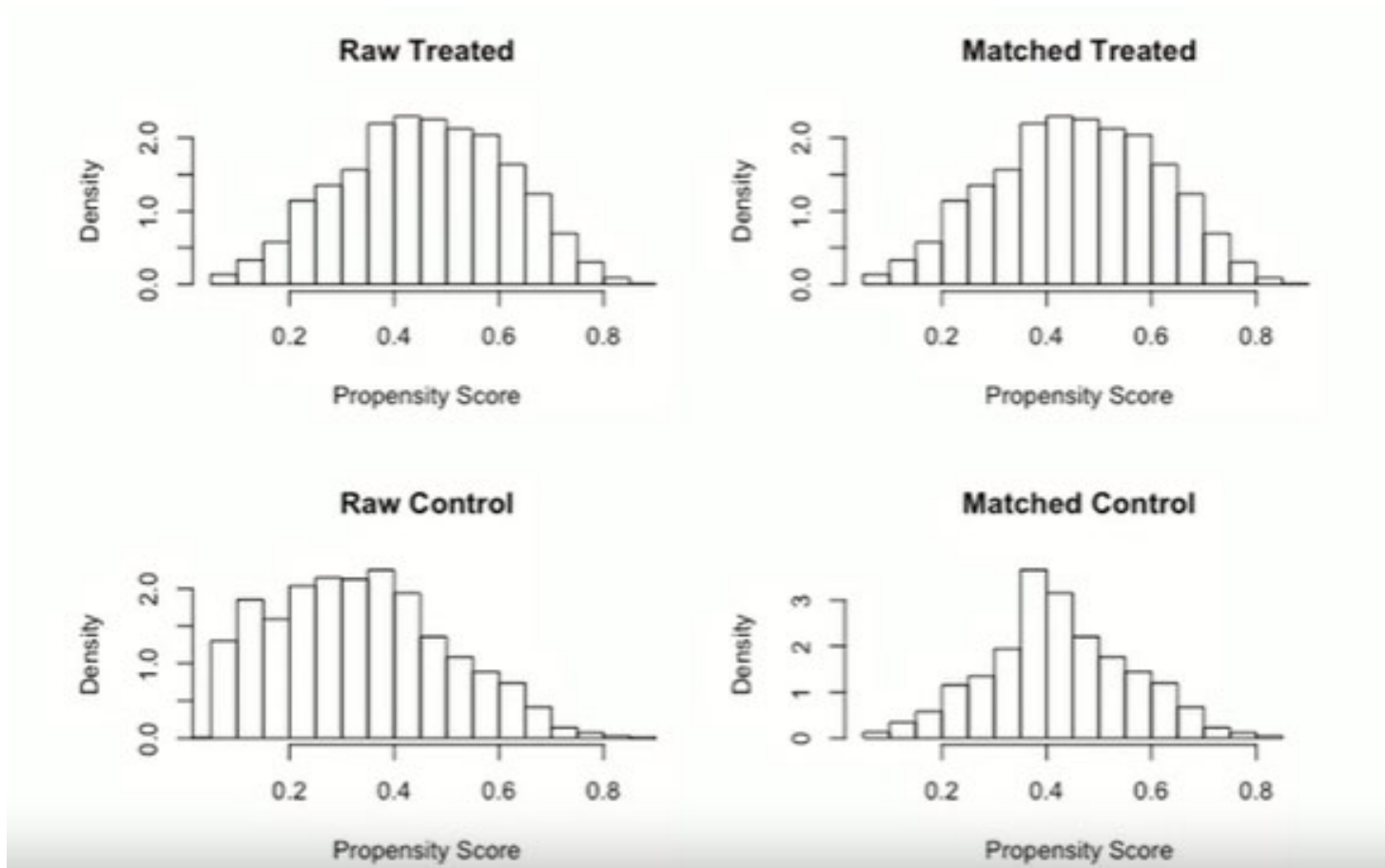
\* Other methods for estimating a propensity score could be used, including machine learning methods

## Example in R: Greedy matching with caliper

```
#re-do matching using a caliper

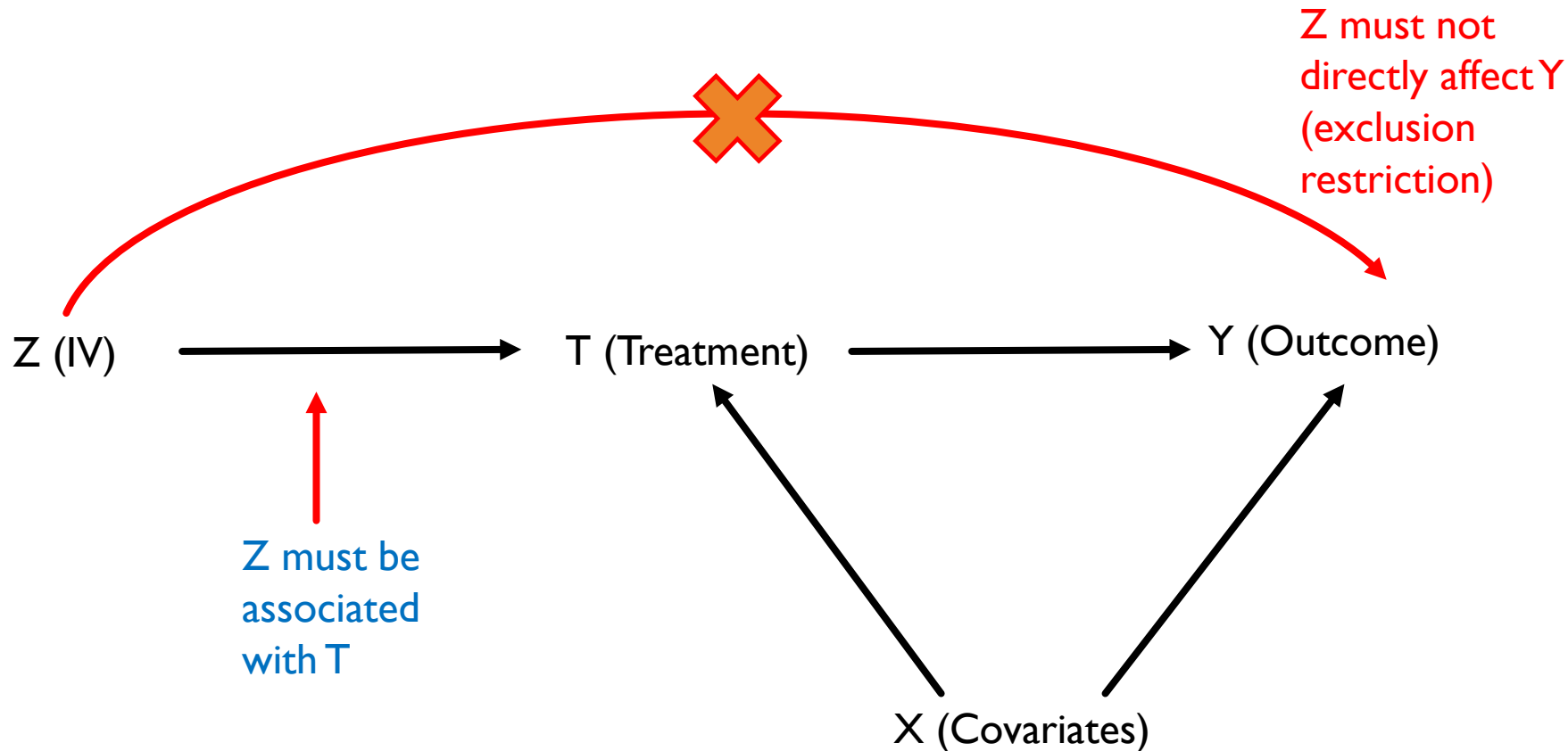
psmatch<-Match(Tr=mydata$treatment,M=1,X=logit(pscore),replace=FALSE,caliper=.2)
matched<-mydata[unlist(psmatch[c("index.treated","index.control")] ), ]
xvars<-c("ARF","CHF","Cirr","colcan","Coma","lungcan","MOSF","sepsis",
         "age","female","meanbp1","aps")
matchedtab1<-CreateTableOne(vars=xvars, strata ="treatment",
                           data=matched, test = FALSE)
print(matchedtab1, smd = TRUE)
```

# Example: Propensity score distribution before/after matching



## Instrumental variables: Exclusion restriction (1/2)

- Instrumental variables (IV) is an **alternative causal inference method** that **does not rely on the ignorability assumption**
- With proper IV, unobserved confounder problem can be handled

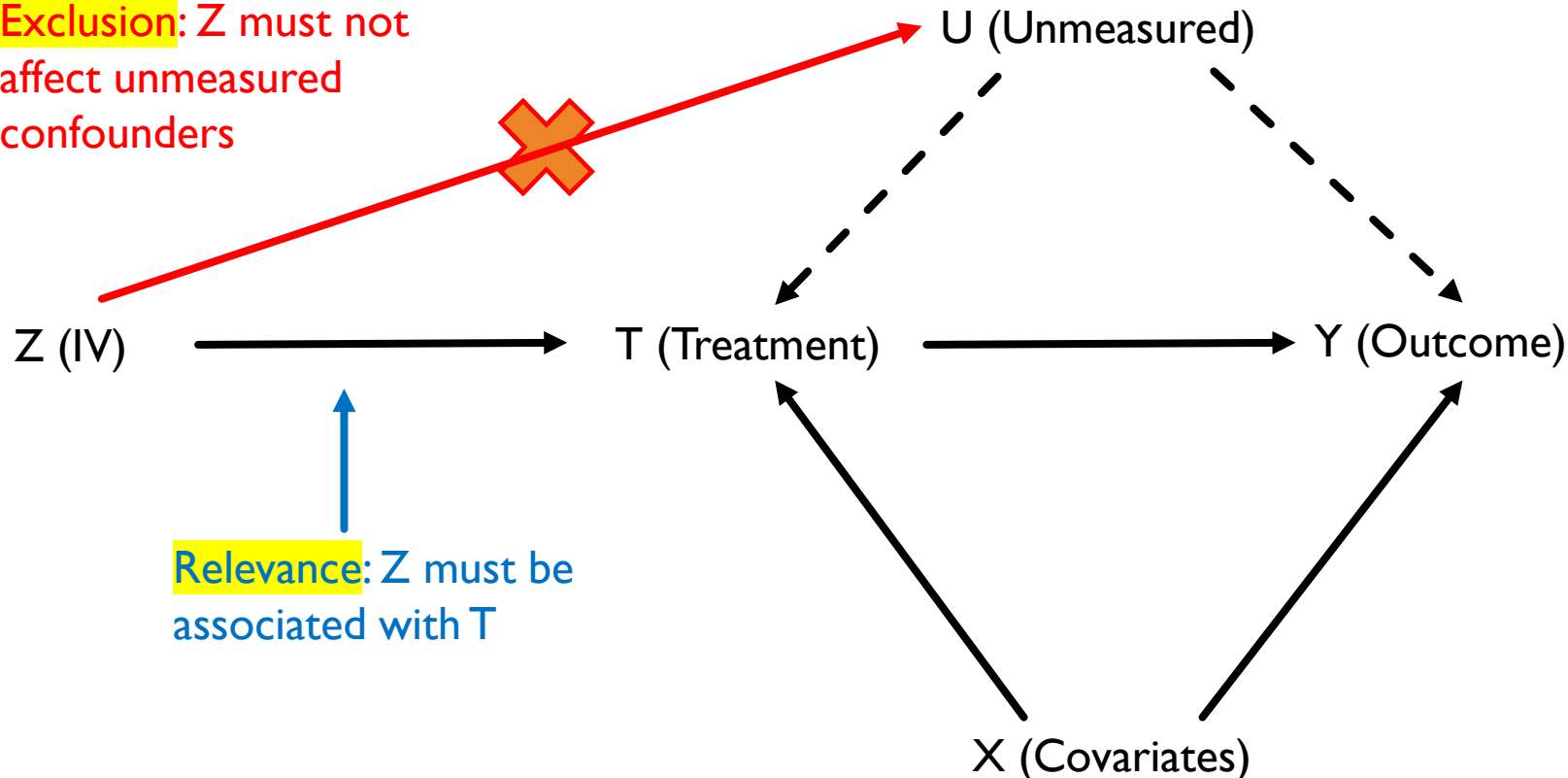


- Z affects treatment, but does not directly, or indirectly through its effect on U, affect the outcome
  - **Highly correlated with T**: can be checked with data
  - **Not correlated with unobserved confounders**: can't be checked with data – logical reasoning
- Think of Z as **encouragement** (randomizer)

## Instrumental variables: Exclusion restriction (2/2)

- Instrumental variables (IV) is an **alternative causal inference method** that **does not rely on the ignorability assumption**
- With proper IV, unobserved confounder problem can be handled

**Exclusion:** Z must not affect unmeasured confounders



**Relevance:** Z must be associated with T

- Z affects treatment, but does not directly, or indirectly through its effect on U, affect the outcome
  - **Highly correlated with T:** can be checked with data
  - **Not correlated with unobserved confounders:** can't be checked with data – logical reasoning
- Think of Z as **encouragement** (randomizer)

## Instrumental variables: Controlled Experiment Example

- T: smoking during pregnancy (Yes/No)
- Y: birthweight
- X: parity, mother's age, weight, etc.
- **Concern:** Could be unmeasured confounders
- **Challenge:** not ethical to randomly assign smoking to pregnant women
- An **intention-to-treat (ITT)** analysis would focus on **the causal effect of encouragement**
  - $E(Y^{Z=1}) - E(Y^{Z=0})$
- **IV methods** can say about **causal effect of smoking** itself (for **compliers**)
- IV: can be randomly assigned or believed to be randomized in nature (natural experiment)
- Z: randomize to **either receive encouragement to stop smoking ( $Z=1$ ) or receive usual care ( $Z=0$ )**
- Sexton and Hebel (1984) "A clinical trial of change in maternal smoking and its effect on birth weight", JAMA

## Instrumental variables: Example – Distance as IV

- Distance to a specialty care center has been used as an IV for the effect of specialty care on health outcomes
  - Shorter distance: stronger encouragement
  - Is distance likely associated with outcomes?
- Z: differential travel time from nearest high-level NICU to nearest regular hospital
- T: delivery at high level NICU vs. regular hospital
- Y: mortality
- Baiocchi et al. (2010) “Building a Stronger Instrument in an Observational Study of Perinatal Care for Premature Infants”



# Instrumental variables: Estimation – 2SLS (2 Stage Least Squares)

- Suppose that we have treatment  $T$  and outcome  $Y$  and model:

- $Y_i = \beta_0 + \beta_1 T_i + \varepsilon_i$

- Usual assumption: error term and covariate ( $T$ ) are independent
- Confounding with unobserved covariate:  $T_i$  and  $\varepsilon_i$  will be **correlated**
  - **OLS fails here**
- Two stage least squares is a method for estimating a causal effect in the instrumental variables setting.
- We are assuming  $Z$  is a valid IV
  - Affects  $T$ : **strong correlation** with  $T$  (weak instrument problem with weak correlation)
  - **Exclusion restriction**: correlation of  $Z$  with  $\varepsilon_i = 0$

## 2SLS: Stage I – Purification: Isolate only the T that was caused by Z

- Regress treatment received, T, on the instrumental variable Z
  - $T_i = \alpha_0 + \alpha_1 Z_i + \omega_i$
  - $Z_i$  and  $\omega_i$  are independent
  - Error term: mean 0, constant variance
- Obtain  $\hat{T}_i = \hat{\alpha}_0 + \hat{\alpha}_1 Z_i$  for each unit (Standard OLS)
  - This is the predicted value of T given Z

## 2SLS: Stage 2 - Measurement

- Regress the outcome,  $Y$  on the fitted value from Stage 1,  $\hat{T}_i$ 
  - $Y_i = \beta_0 + \beta_1 \hat{T}_i + \varepsilon_i$
  - By exclusion restriction,  $Z$  is independent of  $Y$ , given  $T$ 
    - $\hat{T}_i$  is projection of  $T$  onto space spanned by  $Z$  ( $\hat{T}_i$ : independent of  $Y$ )
    - Purified  $T$  is uncontaminated by hidden confounders
  - The estimate  $\beta_1$  is the estimate of the causal effect of  $T$
- The 2SLS estimator is a consistent estimator of the **compiler average causal effect**

$$\beta_1 = CACE = \frac{E(Y|Z = 1) - E(Y|Z = 0)}{E(T|Z = 1) - E(T|Z = 0)}$$

## 2SLS more generally

- Stage 1: Regress  $T$  on  $Z$  and covariates  $X$ 
  - Obtain fitted value of  $T$ :  $\hat{T}$
- Stage 2: Regress  $Y$  on  $\hat{T}$  and covariates  $X$ 
  - Coefficient of  $\hat{T}$  is the causal effect
- Sensitivity analysis methods have been developed for each of the IV assumptions
  - Exclusion restriction: If  $Z$  does directly affect  $Y$  by an amount  $\rho$ , would my conclusions change? (Vary  $\rho$ )
  - Monotonicity: If the proportion of defiers was  $\pi$ , would my conclusions change?

# Instrumental variable: Example

- Card, D. “Using Geographic Variation in College Proximity to Estimate the Return to Schooling.”
  - Z: Indicator that the subject grew up near a 4 year college
  - A: subject’s years of education
  - Y: subject’s income
  - X: covariates such as parents years of education, region of country, age, race, IQ score from a test in HS, etc.
- More schooling is associated with higher income, but is it due to the fact that people with more schooling are different in other ways? (Confounding: observed and unobserved)
  - Solution: proximity to college is an IV

- Sample R code

```
> #two stage least squares
>
> #stage 1: regress A on Z
> s1<-lm(educ12~card.data$nearc4)
> ## get predicted value of A given Z for each subject
> predtx <-predict(s1, type = "response")
> table(predtx)
predtx
0.422152560083588  0.54408183146614
                957                2053
>
> #stage 2: regress Y on predicted value of A
> lm(card.data$lwage~predtx)
```

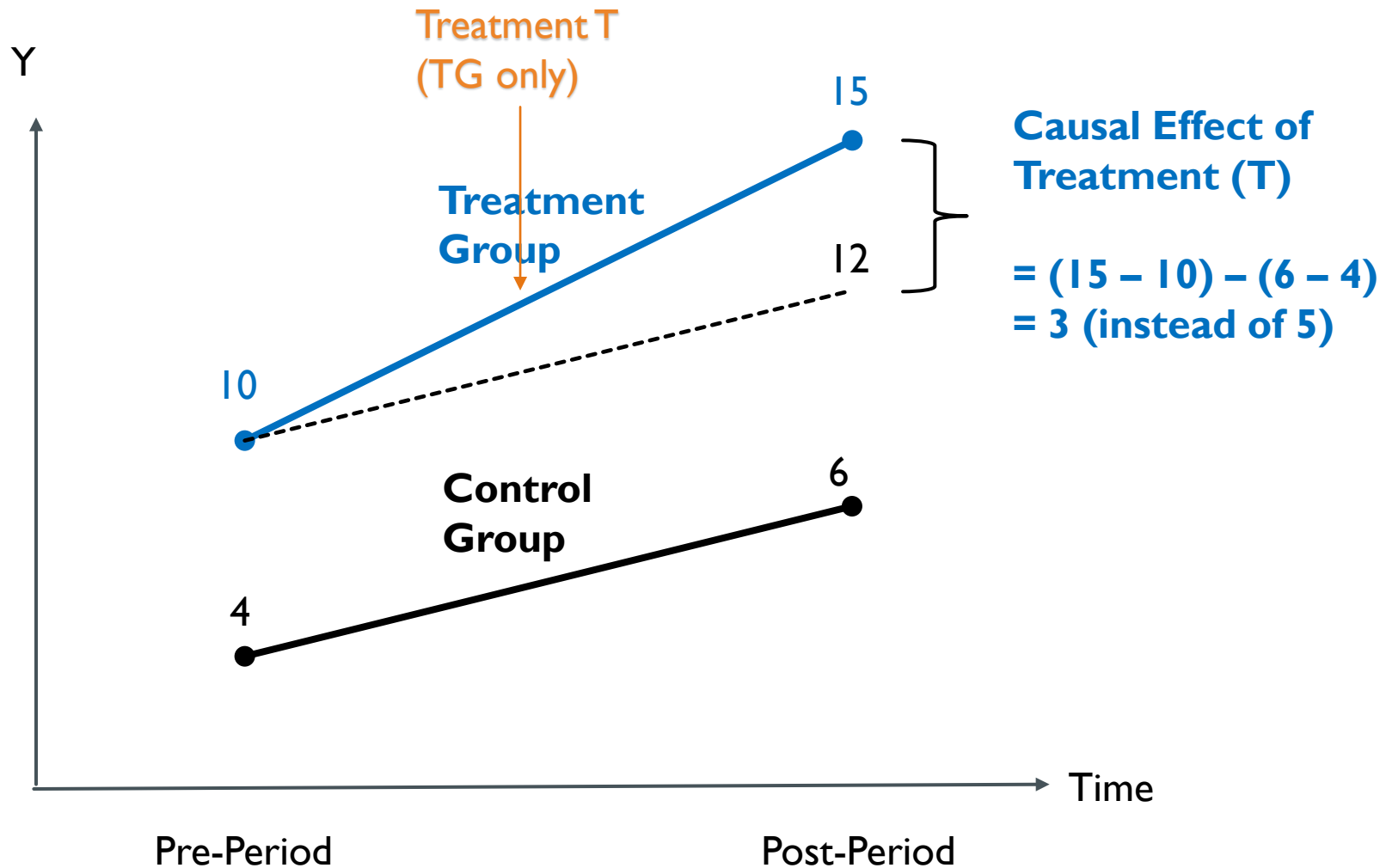
Call:  
lm(formula = card.data\$lwage ~ predtx)

Coefficients:  
(Intercept)      predtx  
          5.616      1.279

# Instrumental variable: Python Example

```
formula = 'log_wage ~ I + C(year_of_birth) + C(state_of_birth) + [years_of_schooling ~ q1+q2+q3]'  
  
iv_many_zs = IV2SLS.from_formula(formula, factor_data).fit()  
  
parse(iv_many_zs)
```

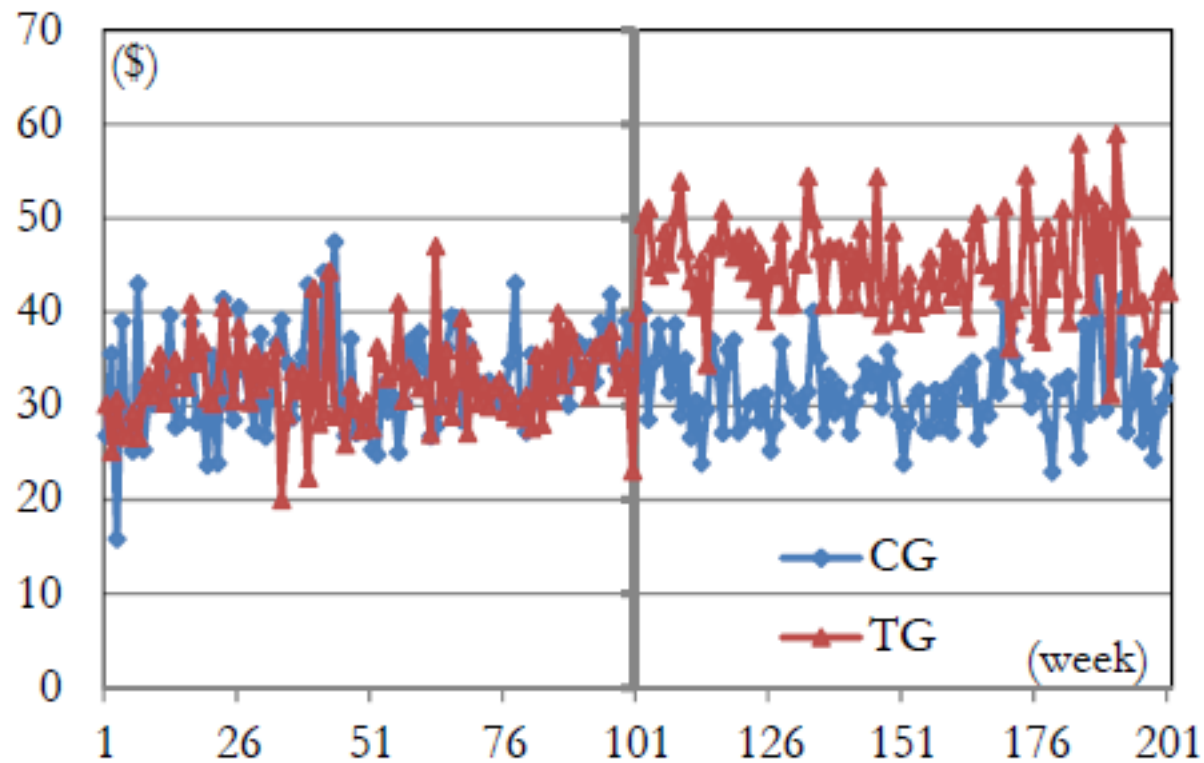
# Difference in differences (DID): Quasi-Experiment



- Key Assumption: [Parallel trend](#)
  - Treatment group and control group might not start out at the same point in Pre-Period
  - You can test this assumption with multiple pre-periods before treatment
- Useful with [natural experiment](#) with [gradual roll-out](#)
- Often combined with “[matching](#)”
- Data requirement: panel data (before/after – treatment/control)

## Parallel Trend Check: Example

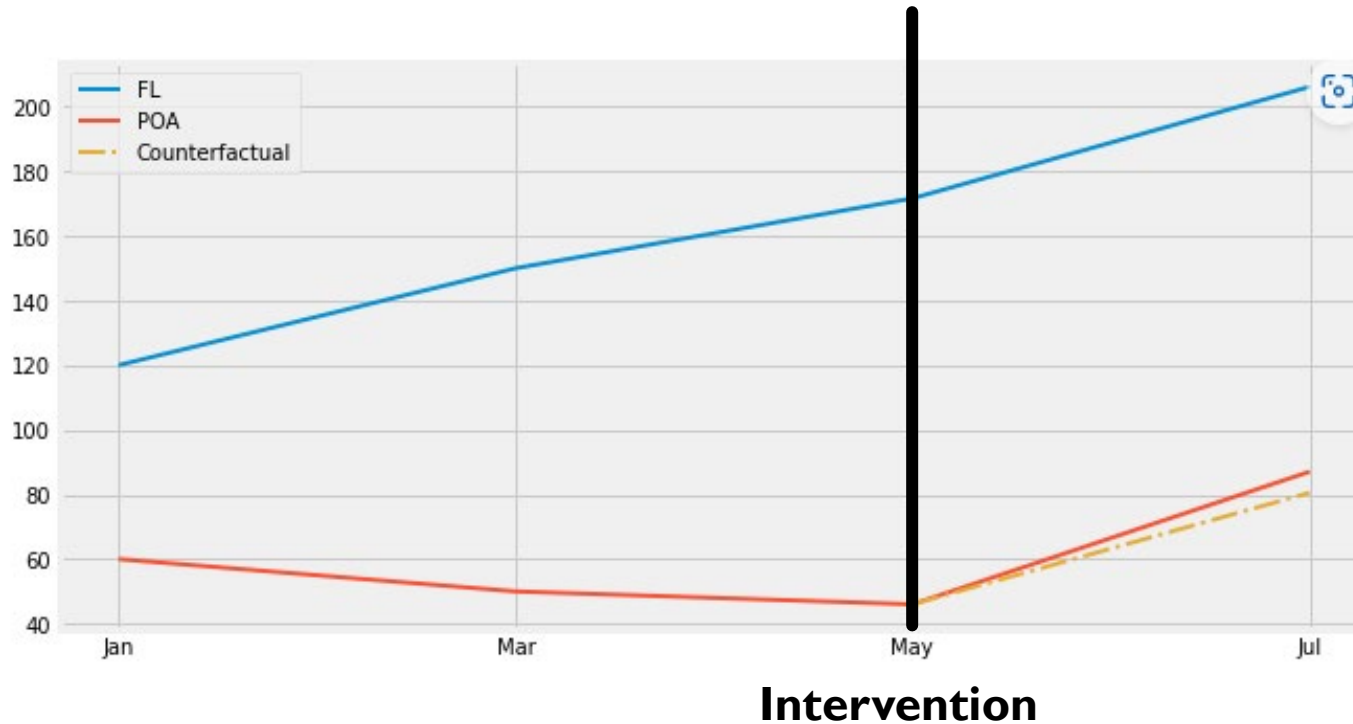
- Weekly Wal-Mart consumer spending before and after supercenter conversion





# Parallel Trend Check: Bad Example

- Pre-periods (before the intervention: billboard advertising) shows **non-parallel trends**
  - Sales in POA: small decreasing trends
  - Sales in FL: steep increasing trends
  - DID: not reliable (biased)
  - Synthetic control can be useful



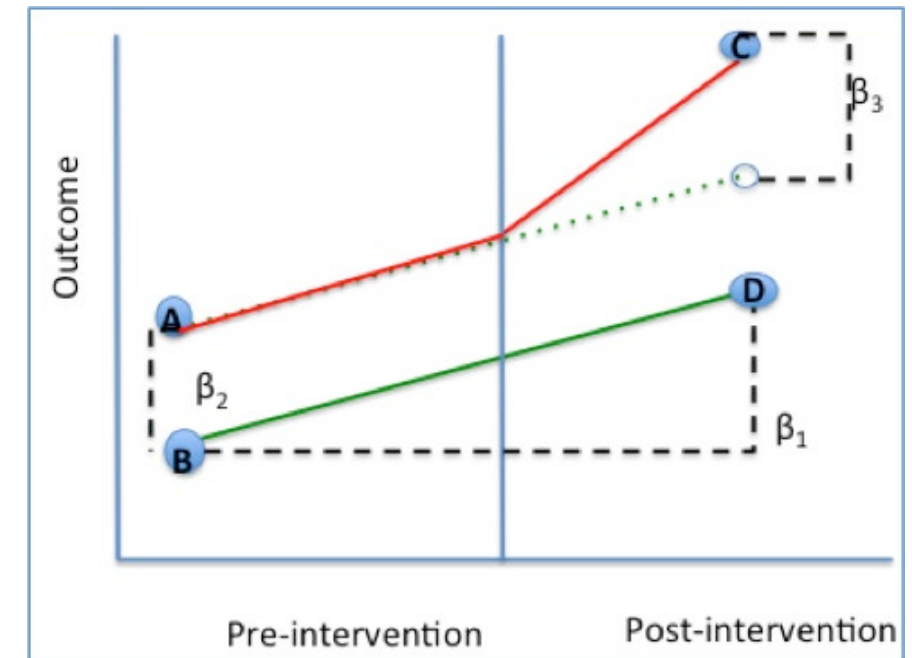
# Difference in differences (DID): Implemented In Regression

- Usually implanted as an interaction term between time (post-intervention dummy) and treatment group dummy variable in a regression model

$$Y = \beta_0 + \beta_1[\text{Time}] + \beta_2[\text{Intervention}] + \beta_3[\text{Time} * \text{Intervention}] + \beta_4[\text{Covariates}] + \varepsilon$$

- More advanced version: Panel DID
  - Multiple treatment and control units
  - Multiple time periods

| Coefficient | Calculation | Interpretation                                 |
|-------------|-------------|------------------------------------------------|
| $\beta_0$   | B           | Baseline average                               |
| $\beta_1$   | D-B         | Time trend in control group                    |
| $\beta_2$   | A-B         | Difference between two groups pre-intervention |
| $\beta_3$   | (C-A)-(D-B) | Difference in changes over time                |



## Difference in differences (DID): Example – Statsmodel in Python

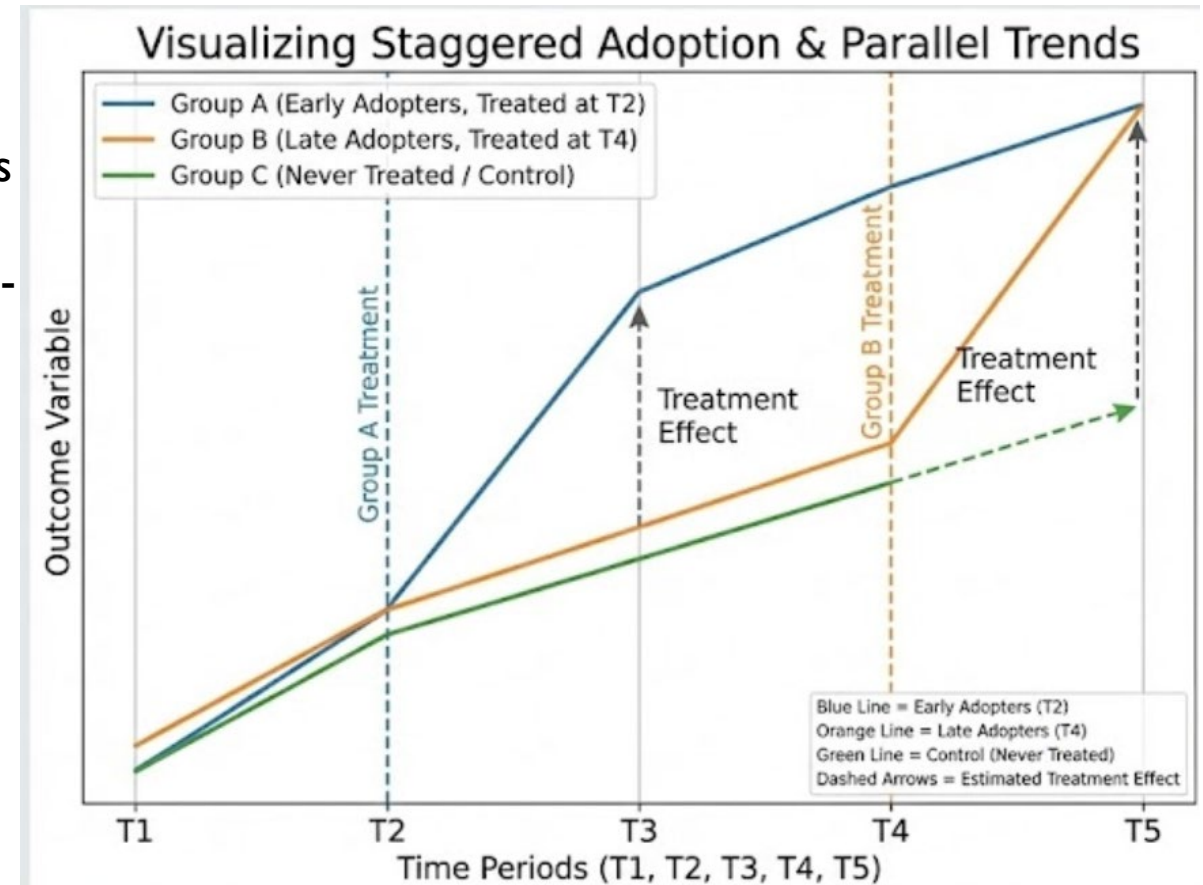
$$Y_i = \beta_0 + \beta_1 POA_i + \beta_2 Jul_i + \beta_3 POA_i * Jul_i + e_i$$

```
smf.ols('deposits ~ poa*jul', data=data).fit().summary().tables[1]
```

|                  | coef      | std err | t       | P> t  | [0.025   | 0.975]   |
|------------------|-----------|---------|---------|-------|----------|----------|
| <b>Intercept</b> | 171.6423  | 2.363   | 72.625  | 0.000 | 167.009  | 176.276  |
| <b>poa</b>       | -125.6263 | 4.484   | -28.015 | 0.000 | -134.418 | -116.835 |
| <b>jul</b>       | 34.5232   | 3.036   | 11.372  | 0.000 | 28.571   | 40.475   |
| <b>poa:jul</b>   | 6.5246    | 5.729   | 1.139   | 0.255 | -4.706   | 17.755   |

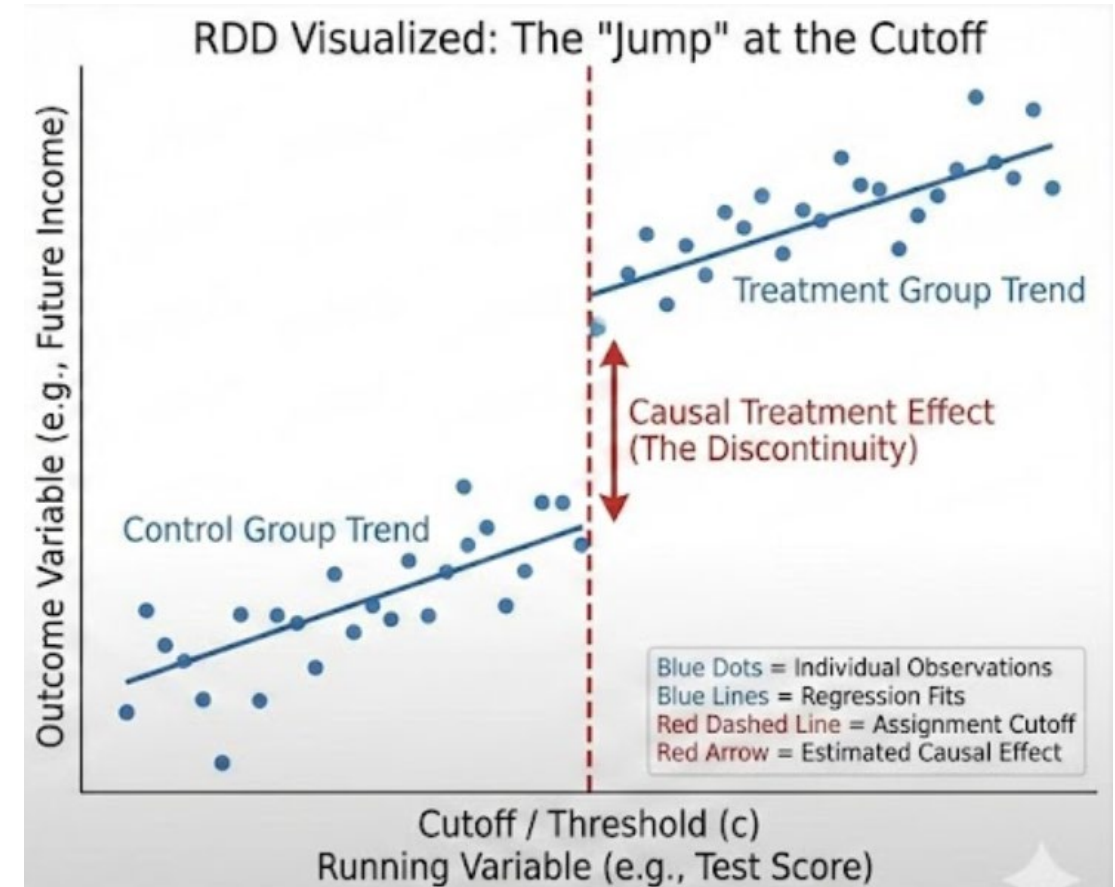
# Advanced Difference in differences (DID): With Staggered Adoption and Multiple Time Periods

- Units adopt treatment at different time periods
- Long-term effects can be different from short-term effects
- Compare changes in outcomes for treated units vs. yet-to-be-treated (or never-treated) units
- Group-Time effects are estimated for each group-time combinations
- Relies on the “parallel trends” assumption
- Allows for estimating treatment effects that vary over time or across cohorts



# Regression Discontinuity Design (RDD)

- Exploits a cutoff on a continuous “running variable” to assign treatment (non-random)
- Compares the units just above and below the threshold (LATE: local average treatment effect)
- Assumption: Units near cutoff are similar, “as-if” randomly assigned
- The “Jump” (discontinuity in outcome) at cutoff indicates causal effect
- Types: Sharp (deterministic assignment) vs. Fuzzy (probabilistic change)



# Double/Debiased/Orthogonal Machine Learning (I/4)

- Double Machine Learning helps with causal inference by acting as a highly **sophisticated "weeding" mechanism**. It clears out **massive amounts of observational background noise (nuisance variables)** so you can actually see the **pure cause-and-effect relationship** underneath
- In causal inference, the ultimate goal is to isolate the effect of  $X$  on  $Y$ . If you are analyzing a pricing experiment, you want to know exactly how a price change ( $X$ ) affects demand ( $Y$ ).
- The **problem** is that  $X$  and  $Y$  are rarely alone. They are surrounded by hundreds of other variables—seasonality, competitor pricing, customer demographics, marketing spend. If you just run a standard regression, the model gets confused by all these overlapping correlations. If you use **standard Machine Learning**, the model tends to **overfit on the background noise** and **"regularizes" (shrinks) your target causal effect toward zero**.



If you combine your predictions in a specific way, the **small ML mistakes will cancel themselves out**.

# Double/Debiased/Orthogonal ML (2/4)

## Frisch-Waugh-Lovell (FWL) Theorem on Steroids

In traditional statistics, the FWL theorem proves that you can find the effect of  $X$  on  $Y$  by first "regressing out" all the background variables from both  $X$  and  $Y$ , and then comparing what is left over.

Double Machine Learning scales this concept up for the modern era. Instead of using simple linear regressions to clear out the background variables, it uses powerful, flexible ML algorithms (like Random Forests or Gradient Boosting).

- **It isolates the variation in  $Y$ :** It removes any changes in demand that were caused by seasonality or demographics.
- **It isolates the variation in  $X$ :** It removes any price changes that were predictably triggered by seasonality or demographics.

What you are left with are the "residuals"—the completely unexpected price changes and the completely unexpected spikes in demand.

## Neyman Orthogonality (Protecting the Estimate)

Because Machine Learning models are highly complex, they naturally make small estimation errors, especially when tuning their hyperparameters. In a standard setup, these small ML errors would bleed directly into your causal estimate, permanently biasing your measurement of how price affects demand.

Double ML is built around the **Neyman Orthogonal Moment Condition**. This is a mathematical guarantee that the causal estimate (the final step) is **"locally insensitive" to the small mistakes made by the ML models in the earlier steps.**

By separating the prediction problem (handled by the ML) from the causal inference problem (handled by the final comparison of residuals), Double ML prevents the ML model's inherent "greediness" and regularization biases from corrupting the final answer. It allows you to leverage the immense predictive power of ML without sacrificing the strict mathematical rigor required for causal measurement.

# Double/Debiased/Orthogonal ML (3/4)

## Recipe

- Step 1: Split the Data – K folds
- Step 2: ML Model 1 – Outcome Model
  - Predicts Outcome (Y) from Control Variables (X).
  - Train Model  $I_{D=1}$  only using treated units (D=1)
  - Train Model  $I_{D=0}$  only using untreated units (D=0)
  - This provides “Outcome Residuals”
- Step 3: ML Model 2 – Propensity Score Model
  - Predicts Treatment (D) from Control Variables (X) – both treated/untreated units
  - This provides “Treatment Residuals”
- Step 4: Causal Effect – Solve for ATE
  - Solve for Causal Effect by Using **Moment Condition with Outcome Residuals and Treatment Residuals**

$$\hat{\theta}^{ATE} = \frac{1}{n} \sum_{k=1}^K \sum_{i \in I_k} \left( \underbrace{\hat{\alpha}^{-k}(D_i, X_i)(Y_i - \hat{\ell}^{-k}(D_i, X_i))}_{\text{Bias correction}} + \underbrace{\hat{\ell}^{-k}(1, X_i) - \hat{\ell}^{-k}(0, X_i)}_{\text{Direct estimate}} \right)$$

Bias correction

Direct estimate

➡ Average doubly robust components across the whole sample

## Applications and Key Ingredients

- Establishes causal relationships in high dimensional data
- Handles complex, non-linear confounding variables
- Generalization of “Doubly Robust” Estimator
- Key ingredients:
  - **Regularization bias**: addressed by **Neyman Orthogonality** – orthogonality score
  - **Overfitting bias**: addressed by **Cross-fitting**

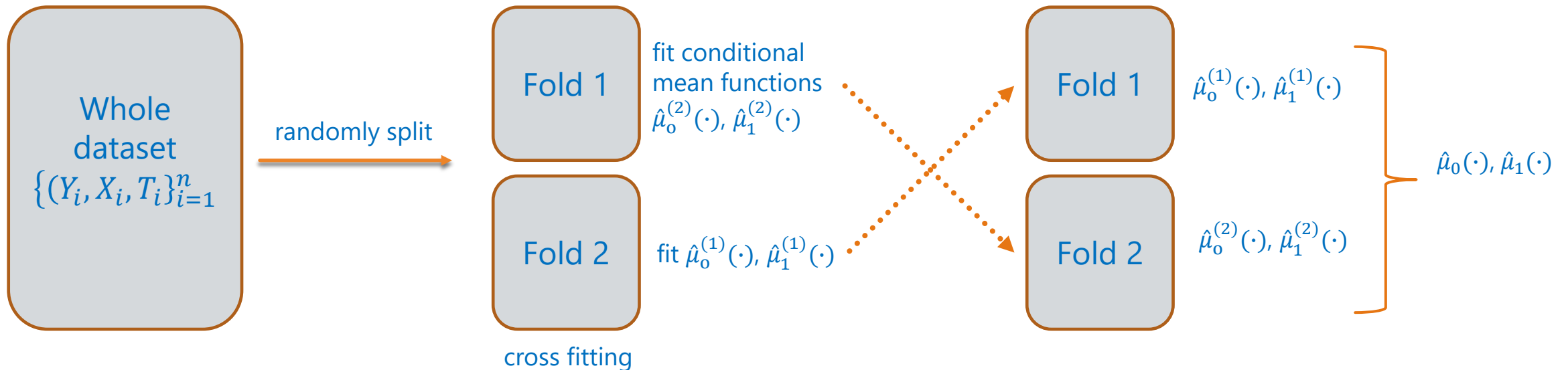
$$\hat{\alpha}^{-k}(D_i, X_i) = \frac{D_i}{\hat{r}^{-k}(X_i)} - \frac{1 - D_i}{1 - \hat{r}^{-k}(X_i)}$$

➡ Weight defined by using the cross-fitted propensity estimates



# Double/Debiased/Orthogonal ML (4/4): Cross-fitting in Outcome Model

- Randomly split the data into K folds: 2 folds work well.
- **Cross-fitting**: For fold  $k$ , use the data in other folds to fit regressors, then apply on the  $X$  in fold  $k$ .
- Any base ML algorithms (e.g., **LightGBM**, Deep Neural Network) can be used to fit **conditional mean functions** with this procedure
- **Debiasing corrections** are applied: ML does not guarantee unbiasedness!



# DML-IV: Double Machine Learning + Instrumental Variable

- (1) The Triple Scrub: Double ML – Background variables are entirely partialled out
  - Predict the Outcome: Get the Surprise Outcome
  - Predict the Treatment: Get the Surprise Treatment
  - Predict the Instrument:
- (2) The IV Purification (Stage 1 of 2SLS) – strict linear model
  - Use the Surprise instrument to predict the Surprise Treatment
  - This isolates only the variation in the treatment that was strictly caused by a clean, random nudge
- (3) The Final Measurement (Stage 2 of 2SLS) – strict linear model
  - Take that newly purified treatment variation and map it against the Surprise Outcome

# Heterogeneous Treatment Effects (HTE): CATE Estimation (1/4)

The **Causal Forest Solution** (GRF with IV)

To measure **Heterogeneous Treatment Effects**—how a treatment affects different people differently—we have to move beyond finding a single "average" number.

If we are testing a new blood pressure medication, DML gives us the Average Treatment Effect (ATE): *"On average, this pill lowers blood pressure by 5 points."*

But a doctor doesn't care about the average. They want the **Conditional Average Treatment Effect (CATE)**: *"How much will this pill lower blood pressure for a 65-year-old woman with a history of smoking?"*

To answer that while still using an Instrumental Variable (IV) to handle hidden confounders, we need a specialized algorithm called a **Causal Forest** (often implemented as a Generalized Random Forest, or GRF).

# Heterogeneous Treatment Effects (HTE): CATE Estimation (2/4)

## Why Standard Random Forests Fail

A standard Random Forest is built for *prediction*. It looks at the data and asks, "*What splits will help me predict the outcome most accurately?*" It might split the data by Age > 50, then by Smoker = Yes, trying to group people with similar blood pressure together.

But we aren't trying to predict blood pressure. We are trying to measure a *causal effect* (the difference the pill makes). You cannot calculate an effect for a single person because you only observe them either taking the pill or not taking it—you never see both.

If you use a standard Random Forest, it will split the data based on who is naturally healthy versus unhealthy, completely ignoring the causal effect of the pill.

# Heterogeneous Treatment Effects (HTE): CATE Estimation (3/4)

## The Causal Forest Solution (GRF with IV)

A Causal Forest fundamentally changes the rules of how the trees are built. It stops optimizing for prediction and starts optimizing for *causal variance*.

### 1. The Setup (Residualization)

Just like in standard DML-IV, we first use machine learning to "scrub" the background noise. We predict the Outcome, the Treatment, and the Instrument using all known covariates, and we save the residuals (the "surprises").

### 2. The "Honest" Splitting Rule

Now we build the Causal Forest using those residuals. Instead of splitting the data to minimize prediction error, the Causal Forest splits the data to maximize the difference in the *treatment effect* between the two new branches.

It uses the **Instrumental Variable math** inside every single split.

**The split question is no longer:** "*Does splitting by Age > 50 help me predict blood pressure?*"

**The split question is now:** "*If I use my Instrument to calculate the 2SLS causal effect, is the effect significantly different for people over 50 versus people under 50?*"

# Heterogeneous Treatment Effects (HTE): CATE Estimation (4/4)

## The Causal Forest Solution (GRF with IV)

A Causal Forest fundamentally changes the rules of how the trees are built. It stops optimizing for prediction and starts optimizing for *causal variance*.

### 3. Cross-Fitting (Honesty)

To prevent the model from hallucinating effects, Causal Forests use a technique called "**honesty**" (which is similar to cross-fitting in DML).

It uses **one half of the data** to decide **where to make the splits** (e.g., deciding that  $\text{Age} > 50$  is a good split). It uses **the other half of the data** to **actually calculate the causal effect** within those newly created leaves.

### The Final Result: Personalized Estimates

Once the forest is built (using hundreds of these specialized trees), you can drop a new patient into the model.

# Other Causal Inference Topics

- Unobserved Heterogeny
  - Random coefficient (Hierarchical, Multi-level) – Bayesian Statistics (Variational Inference) – Continuous Heterogeneity
  - Finite Mixture (Latent Class) - EM Algorithm – Discrete Heterogeneity
  - Fixed Effect – Dummy Variable – Interaction Term
- Heterogeneous Treatment Effect – Meta Learner
  - S-Learner
  - T-Learner
  - X-Lerner
  - Bayesian Additive Regression Tree (BART)
- Discrete Choice Modeling: Utility-based Models
  - Logit
  - Probit
  - Heuristics and Biases: Elimination by Aspect, Consider and Choose
- Time Series and Panel Data
  - Granger Causality
  - Vector Autoregression (VAR): Muti-variate Tme Series – Impulse Response Function
  - Bayesian Structural Time Series: Google Causal Impact Package
  - Marketing Mix Model
  - Spatio-Temporal Models: Kriging, Spatial Econometrics (SAR, SER)
- Synthetic Control: Better Version of Matching

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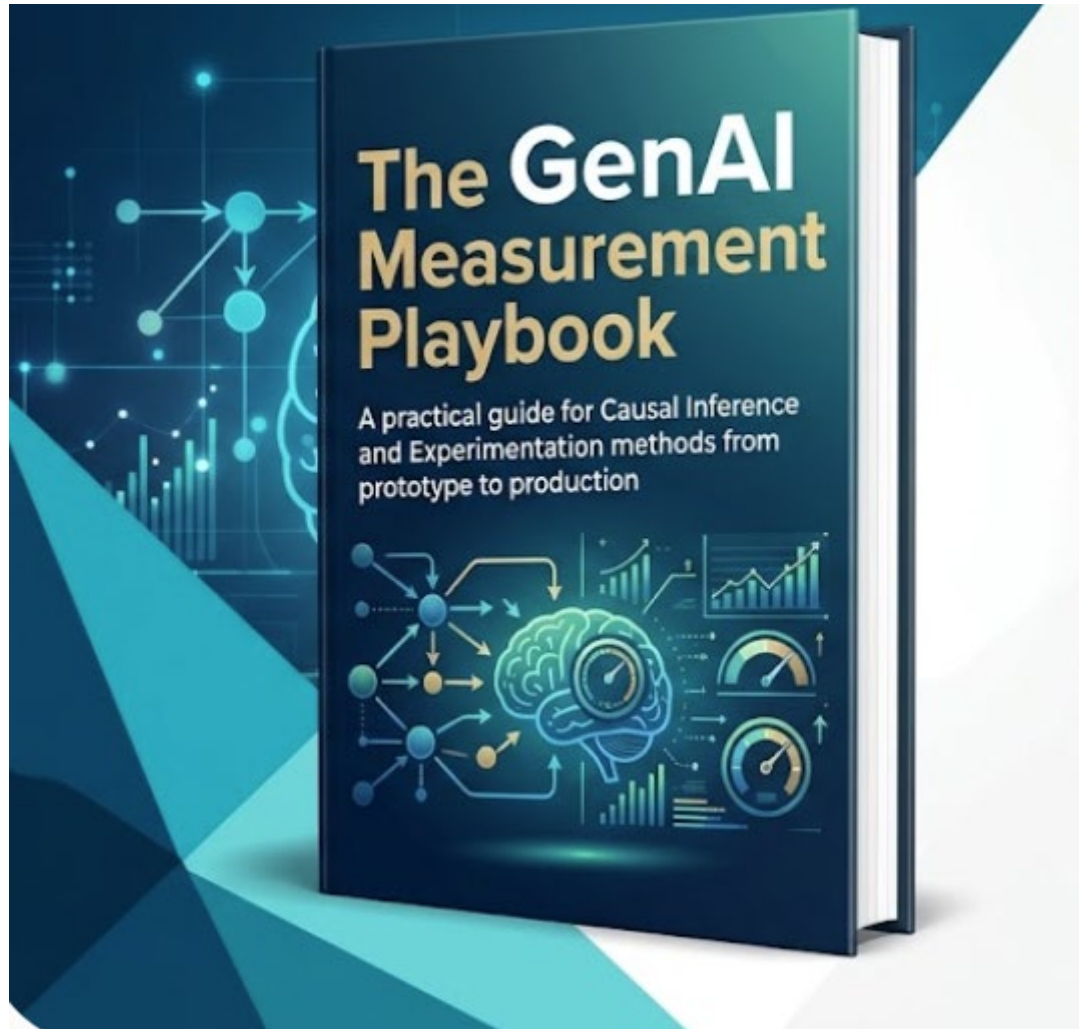
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